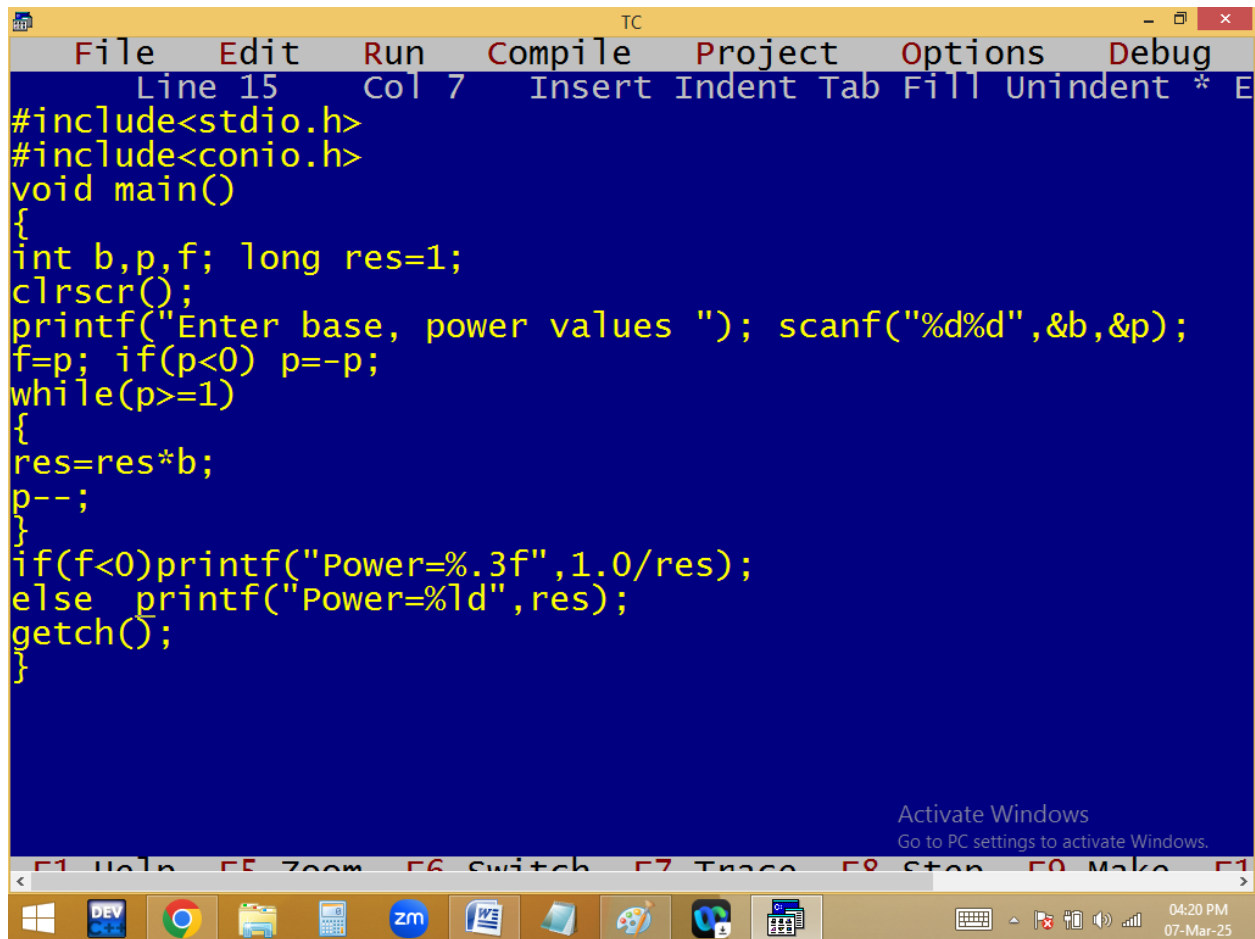


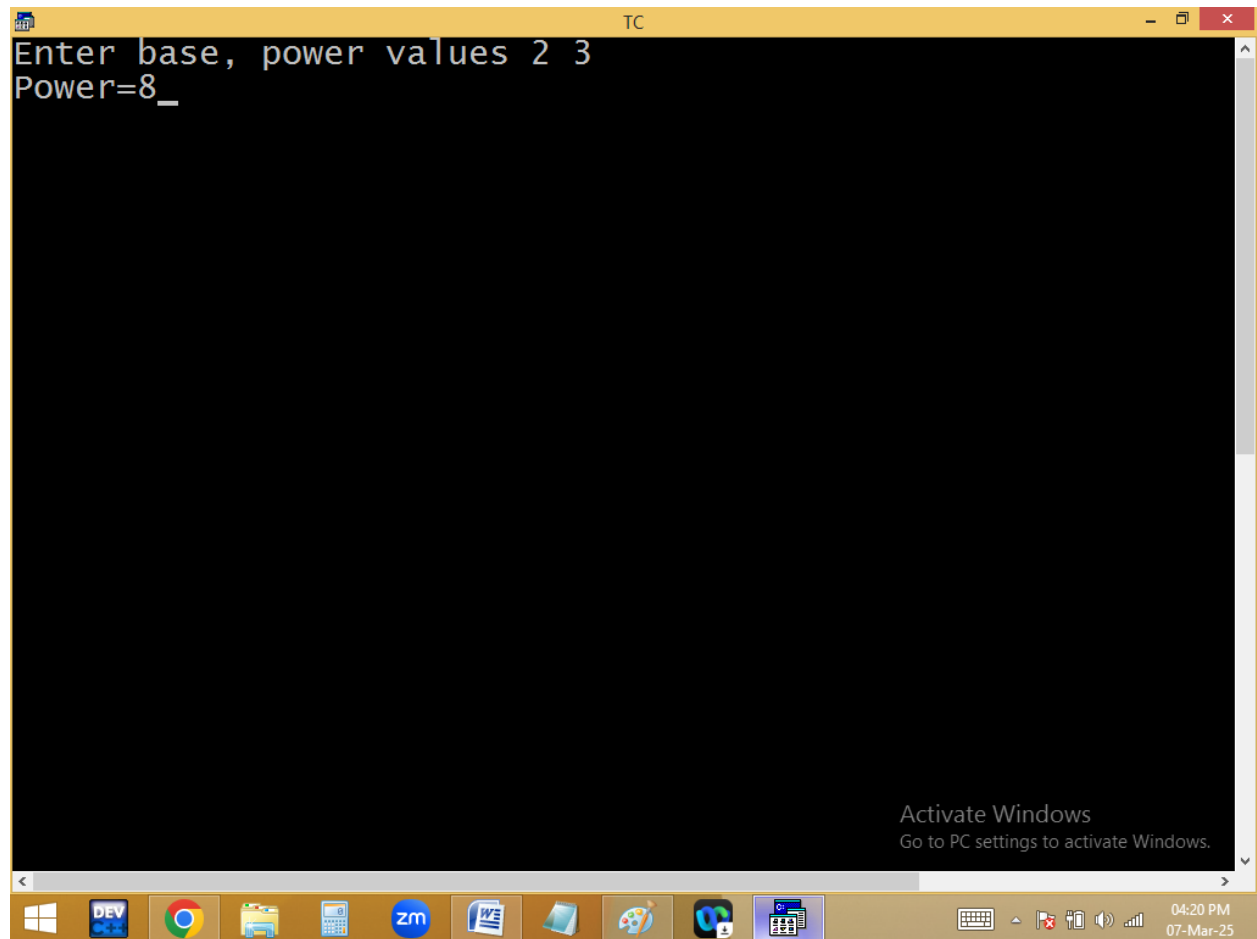


```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p,f; long res=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
f=p; if(p<0) p=-p;
while(p>=1)
{
res=res*b;
p--;
}
if(f<0)printf("Power=%.3f",1.0/res);
else printf("Power=%ld",res);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

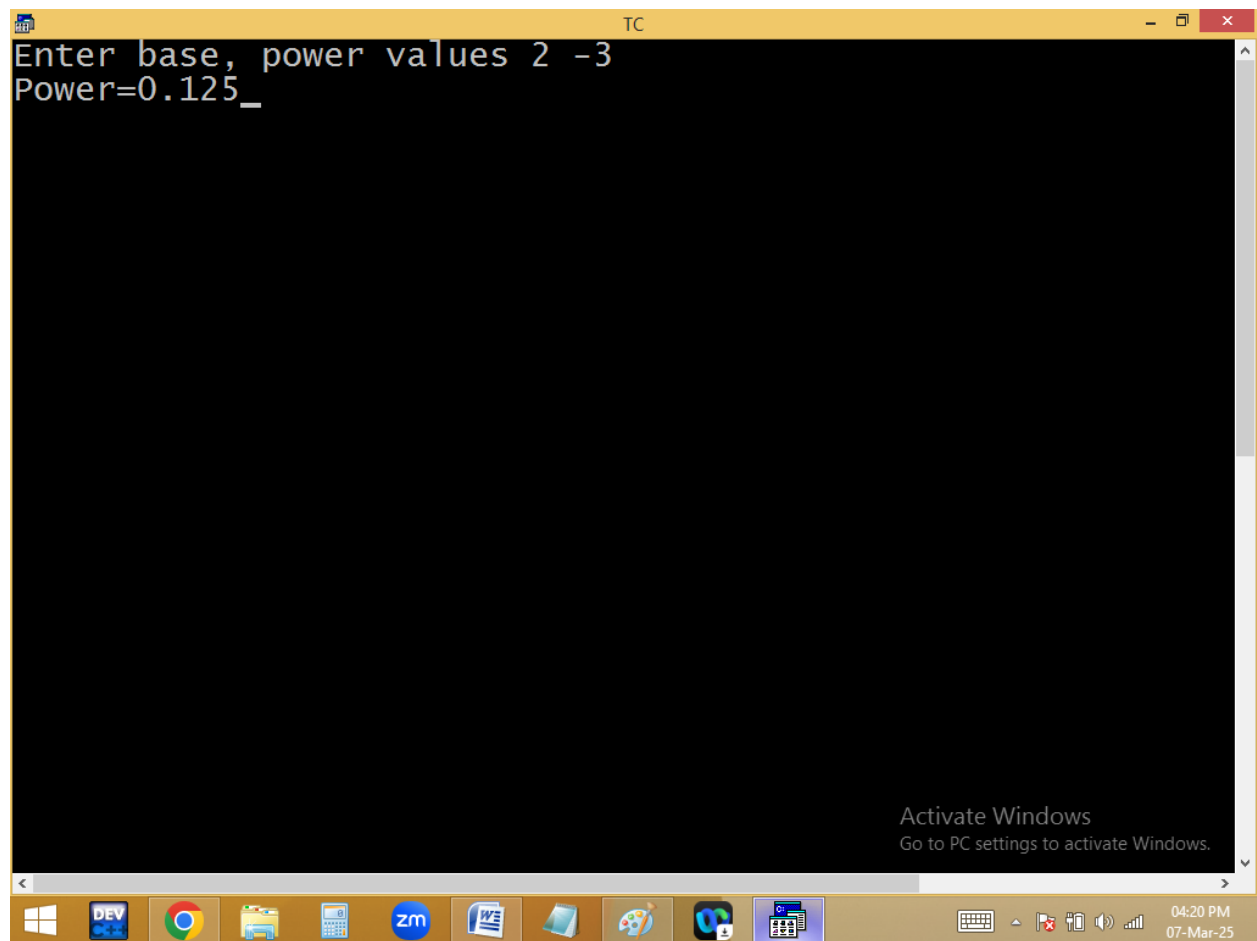
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:20 PM
07-Mar-25

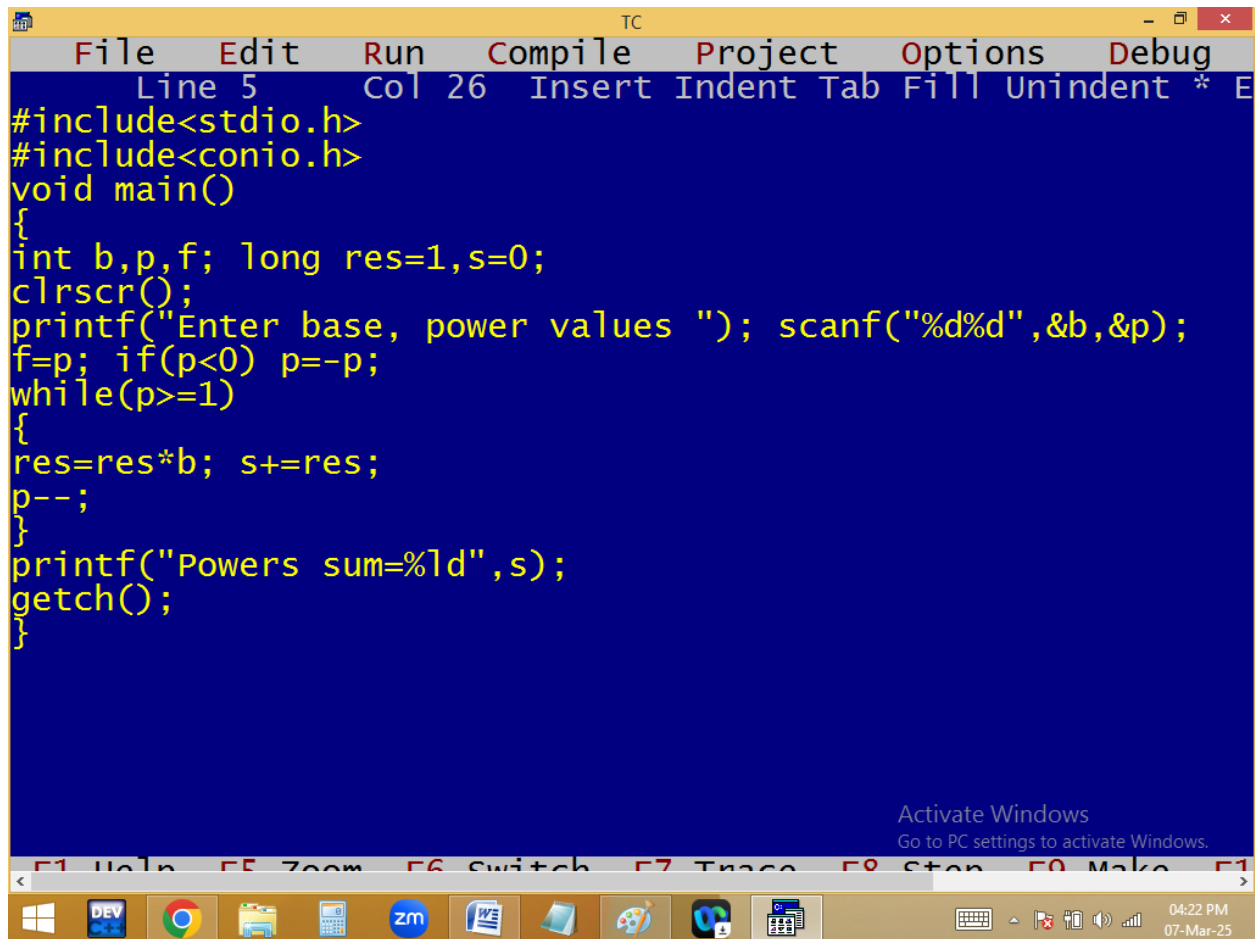


```
TC
Enter base, power values 2 -3
Power=0.125_
```

Activate Windows
Go to PC settings to activate Windows.



Powers sum:

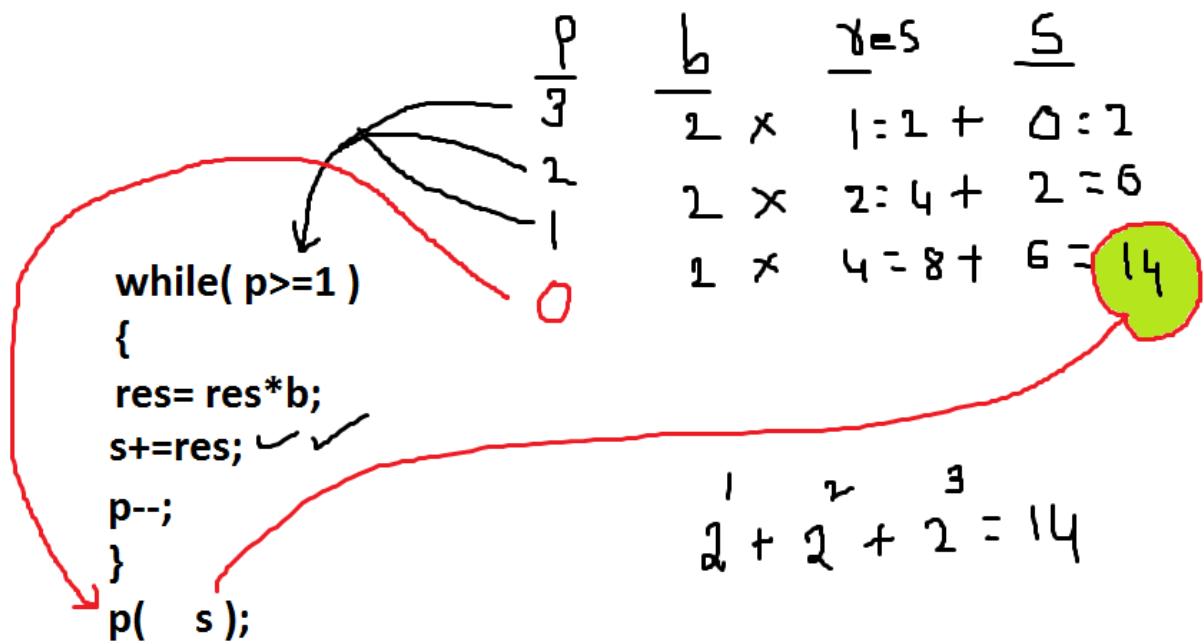
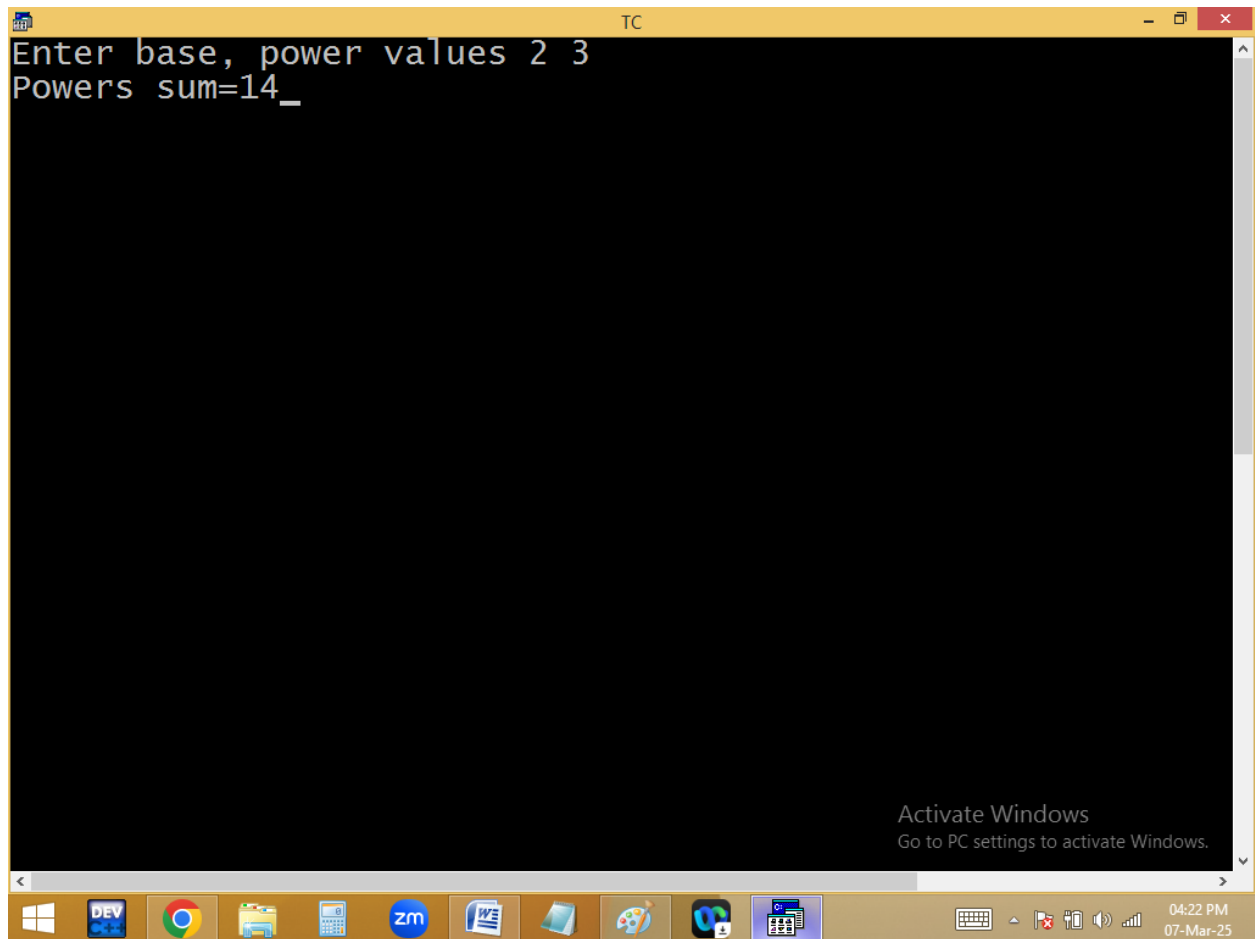


```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 26 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p,f; long res=1,s=0;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
f=p; if(p<0) p=-p;
while(p>=1)
{
res=res*b; s+=res;
p--;
}
printf("Powers sum=%ld",s);
getch();
}
```

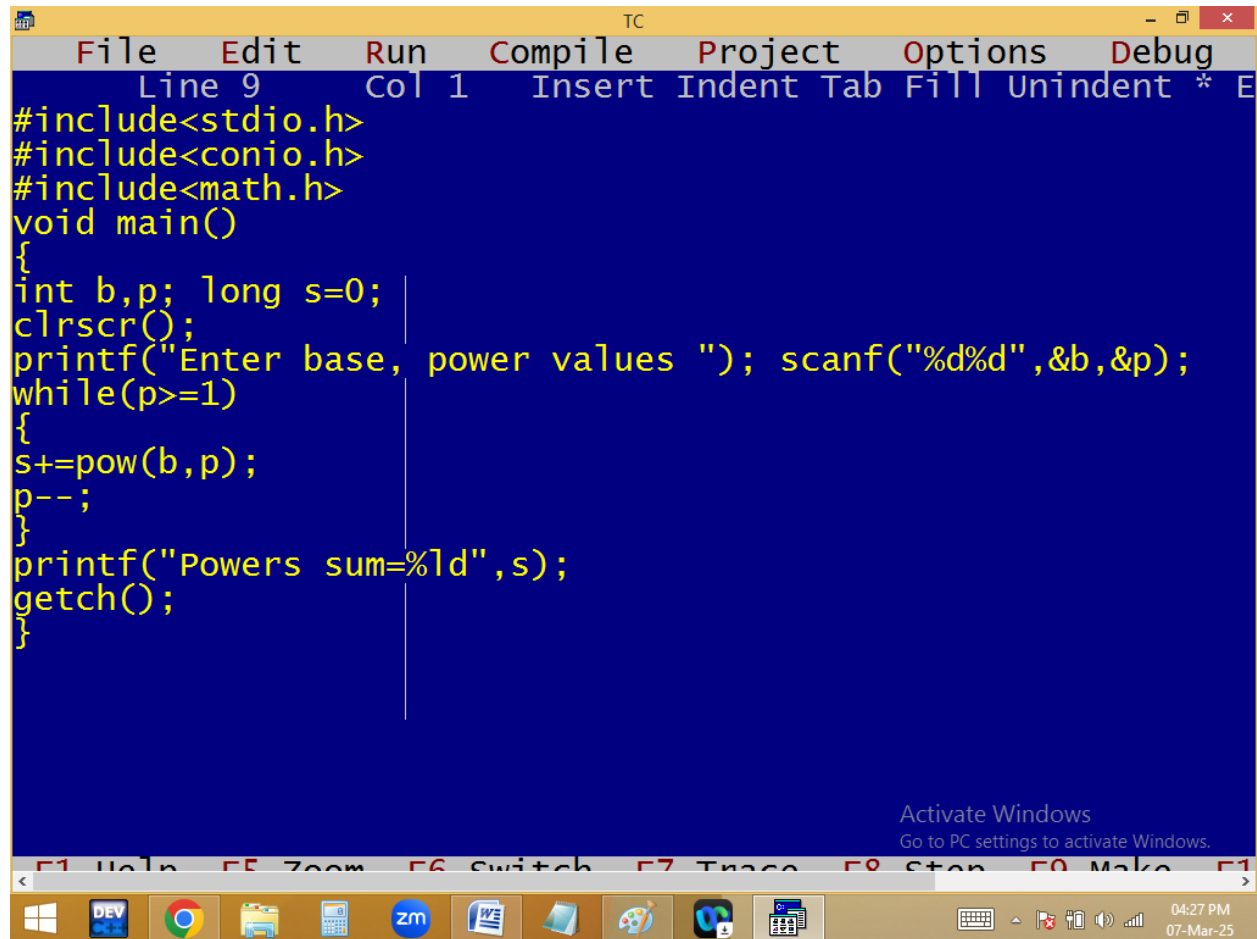
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:22 PM
07-Mar-25



Using pow():

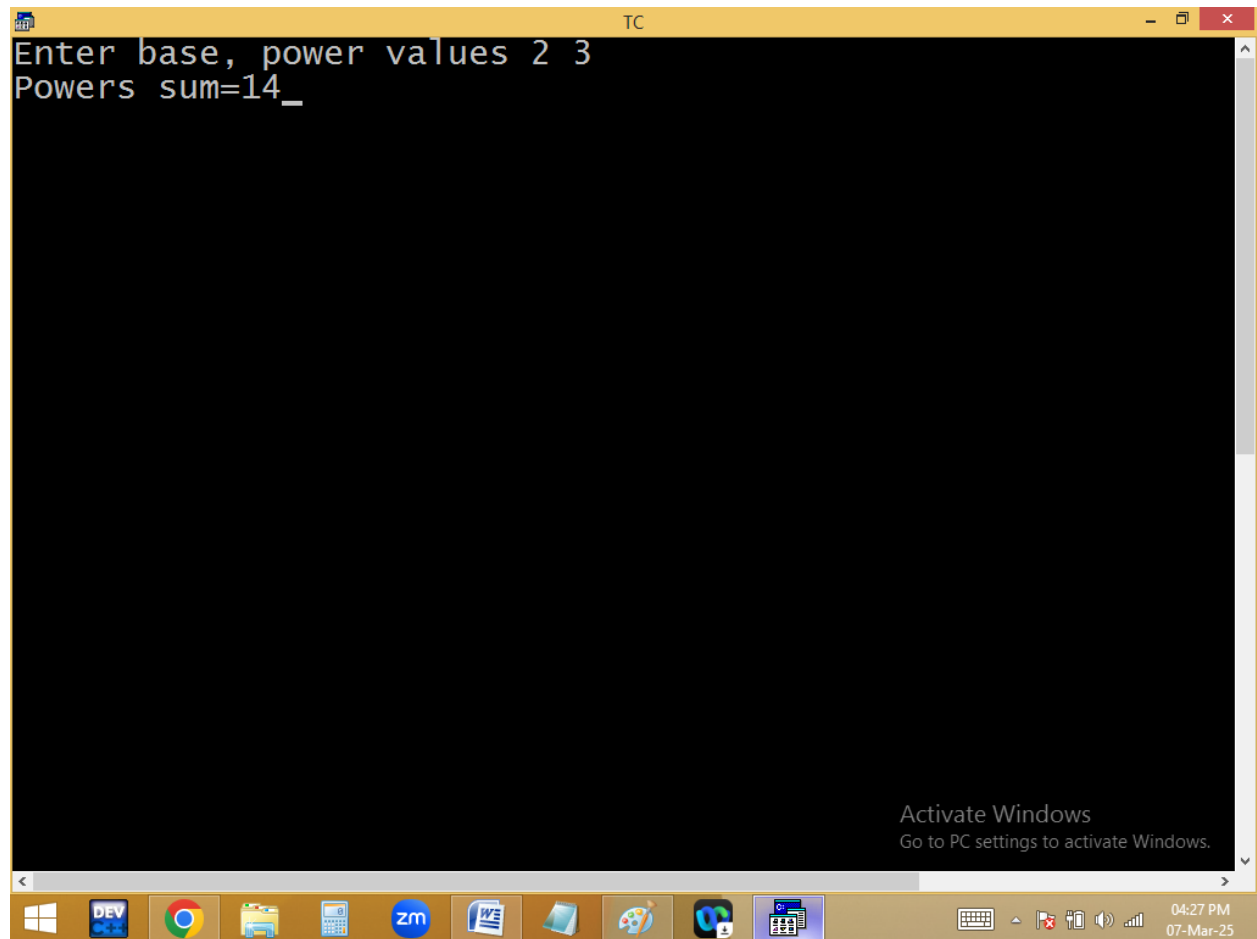


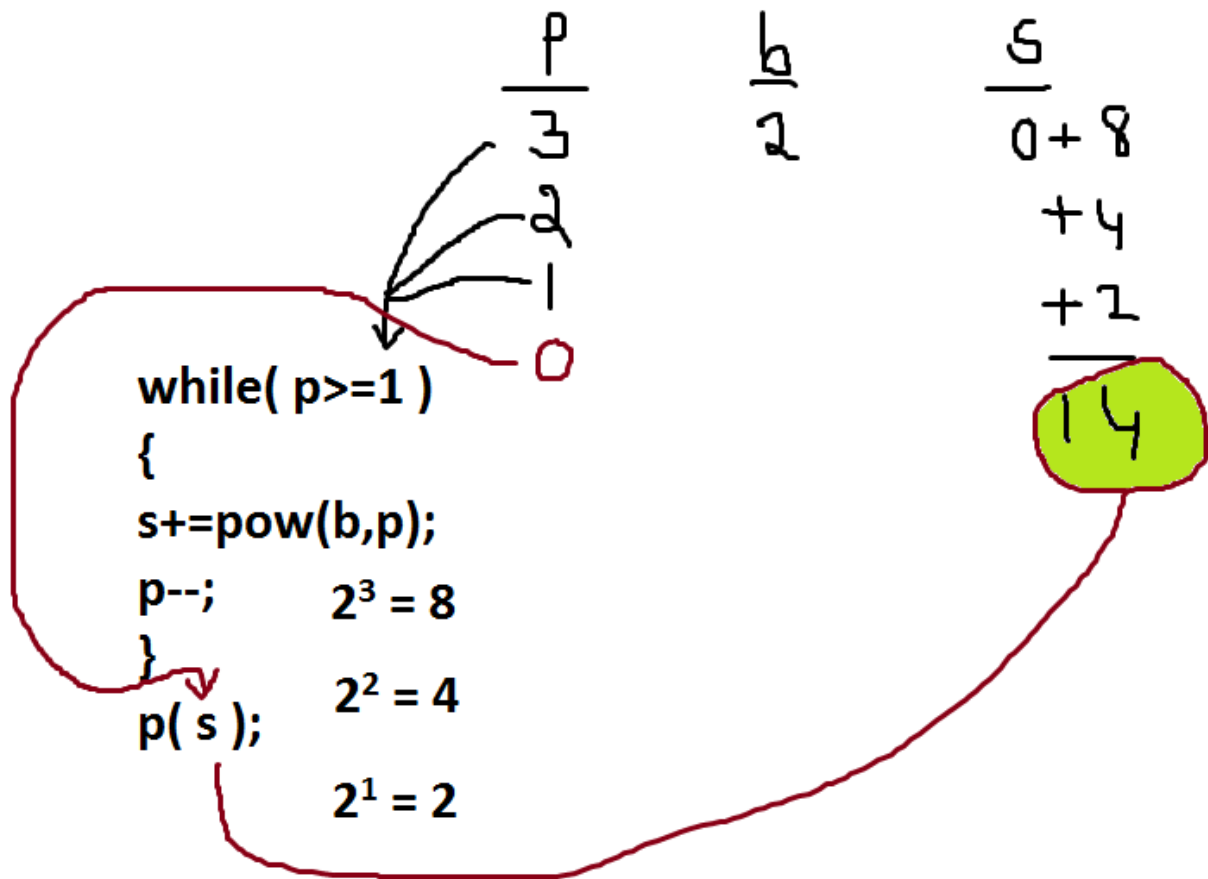
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int b,p; long s=0;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
s+=pow(b,p);
p--;
}
printf("Powers sum=%ld",s);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:27 PM
07-Mar-25



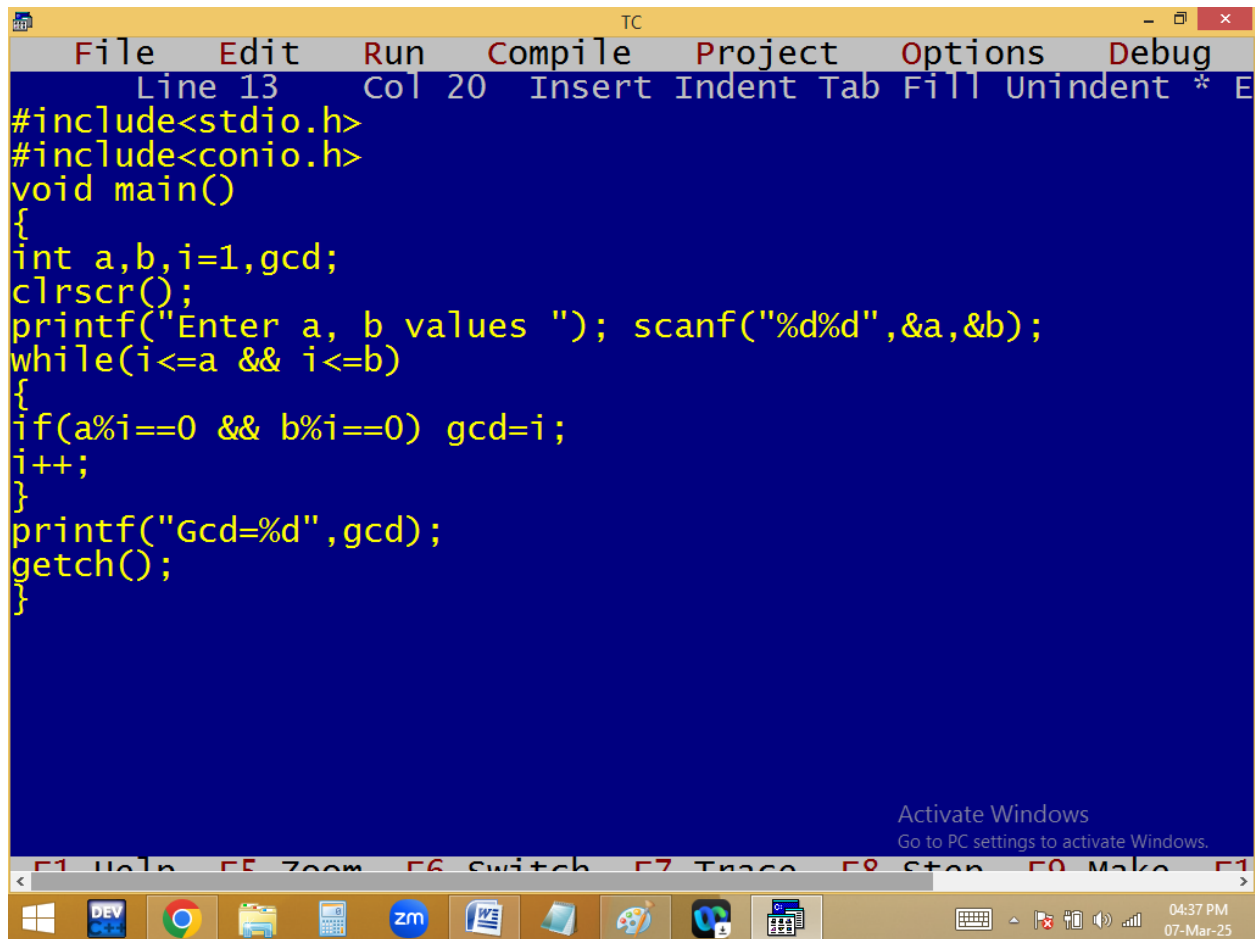


Finding hcf / gcd of given two numbers?

4 factors are 1, 2, 4

6 factors are 1, 2, 3, 6

2 is the gcd of 4 and 6

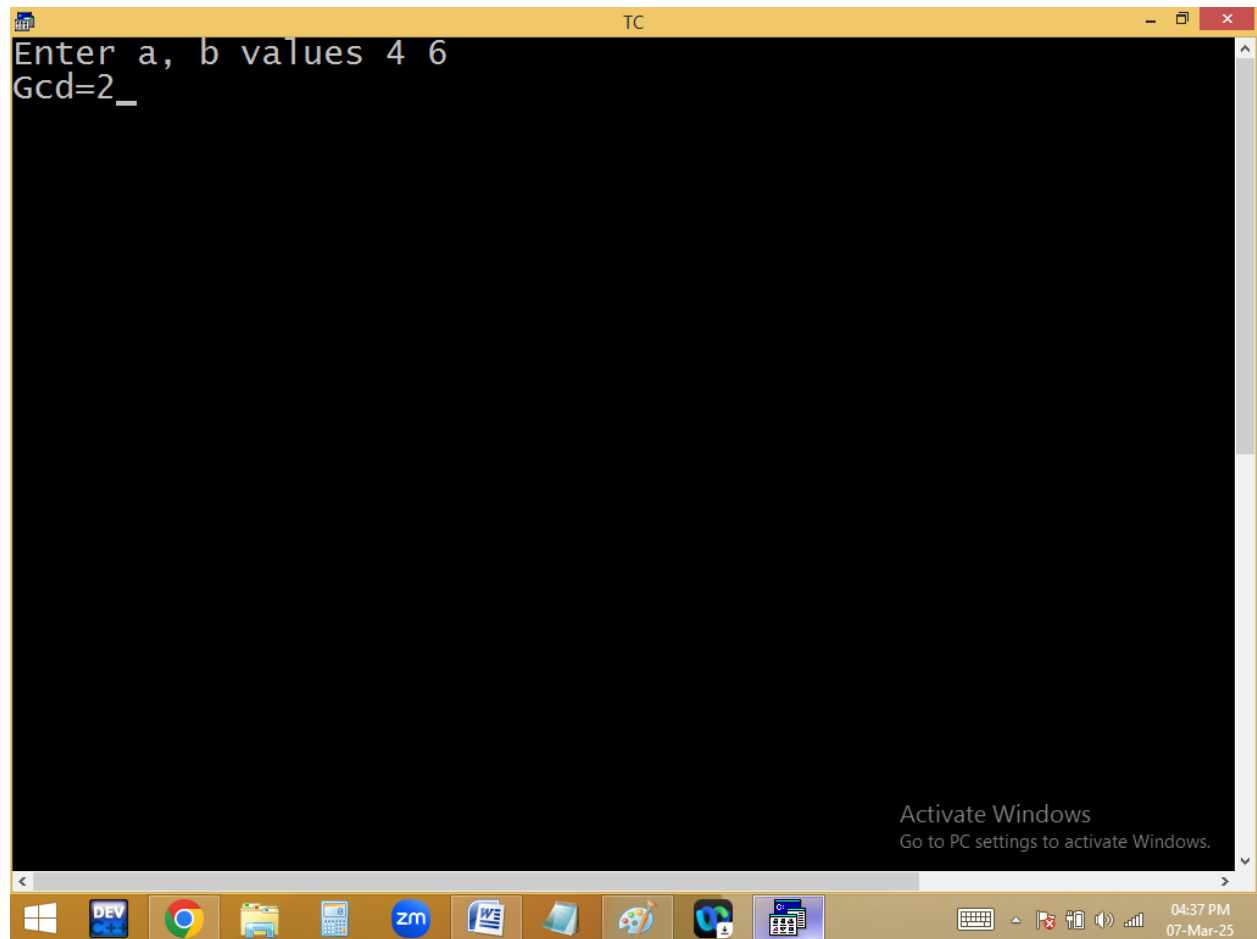


```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 20 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,i=1,gcd;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
while(i<=a && i<=b)
{
if(a%i==0 && b%i==0) gcd=i;
i++;
}
printf("Gcd=%d",gcd);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:37 PM
07-Mar-25

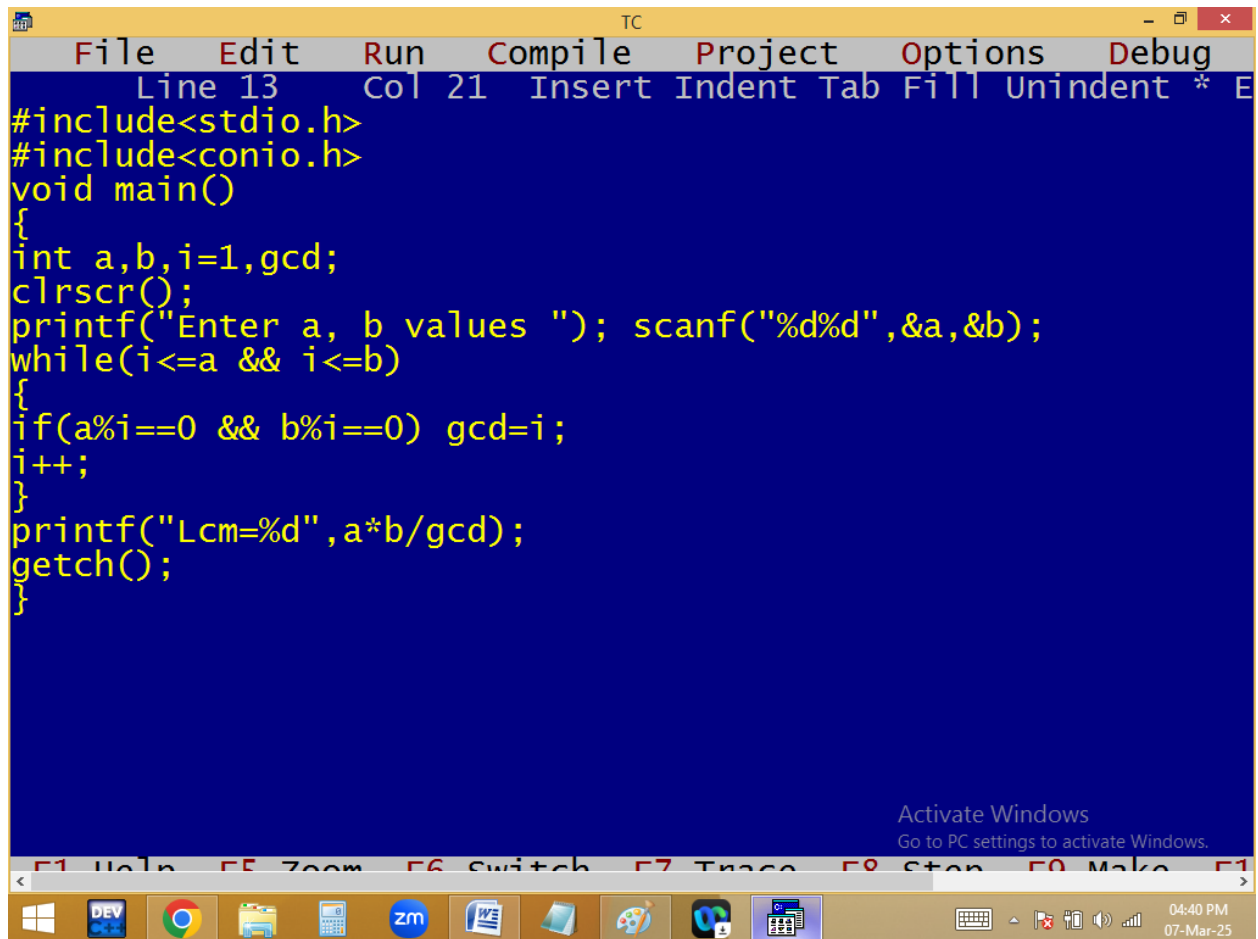


Finding lcm of given two numbers.

Using gcd:

4 and 6 gcd = 2

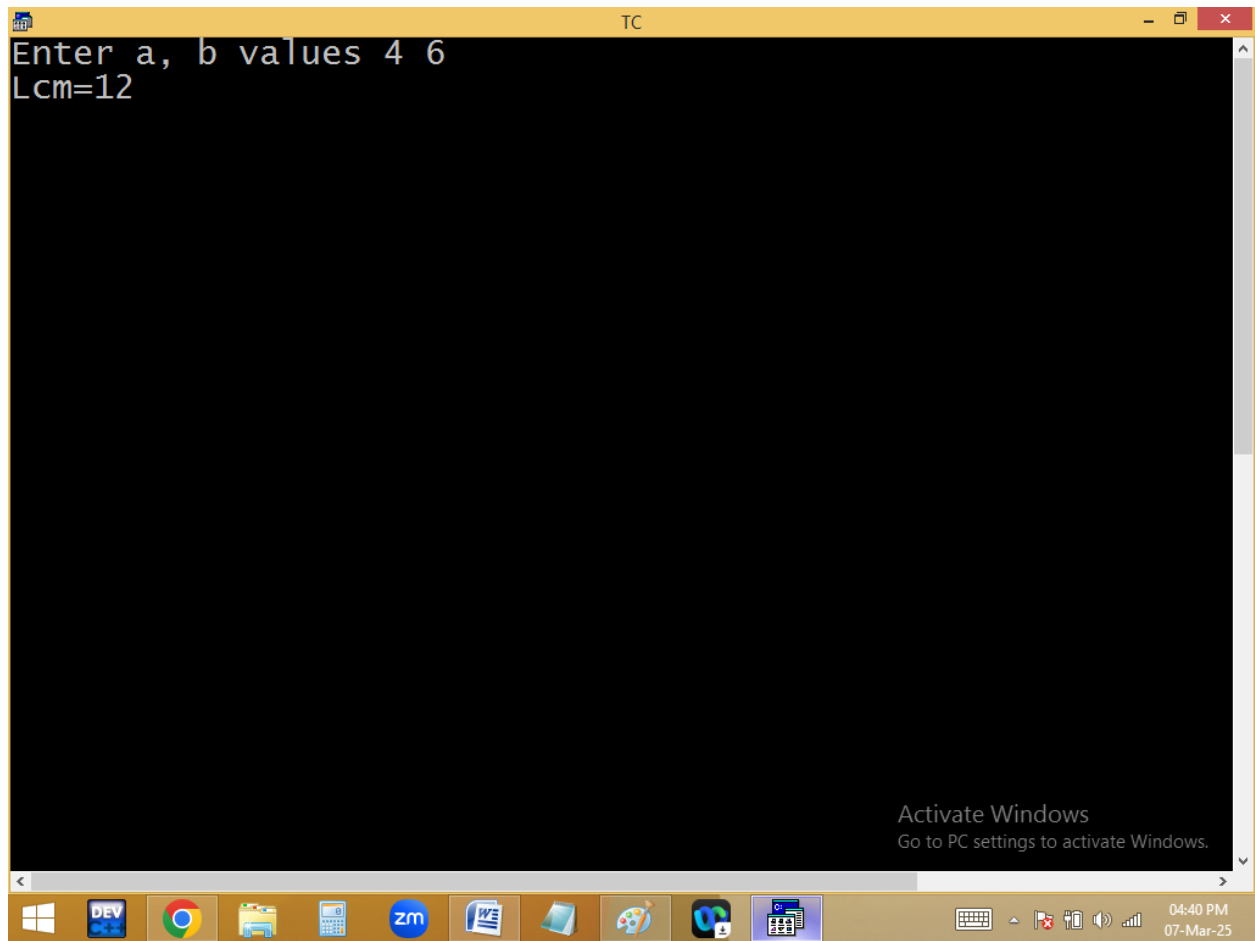
$4 * 6 / 2 = 12$ is lcm



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 21", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,i=1,gcd;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
while(i<=a && i<=b)
{
if(a%i==0 && b%i==0) gcd=i;
i++;
}
printf("Lcm=%d",a*b/gcd);
getch();
}
```

At the bottom of the window, there is a taskbar with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. A system tray on the right shows the time as "04:40 PM" and the date as "07-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



The image shows a screenshot of a Turbo C++ (TC) window. The window title bar is yellow and contains the text "TC". The main area of the window is black and displays the following text in white: "Enter a, b values 4 6" and "Lcm=12". In the bottom right corner of the black area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The window has a standard Windows taskbar at the bottom. The taskbar is light blue and contains several icons: the Windows logo, a folder icon, a Chrome icon, a file explorer icon, a calculator icon, a Zoom icon, a Word icon, a Paint icon, a Teams icon, and a calendar icon. The system tray on the right side of the taskbar shows the time "04:40 PM" and the date "07-Mar-25".

```
Enter a, b values 4 6
Lcm=12
```

Method 2:

max a

$$6 \% 4 = 2$$

$$7 \% 4 = 3$$

$$8 \% 4 = 0$$

$$9 \% 4 = 1$$

$$10 \% 4 = 2$$

$$11 \% 4 = 3$$

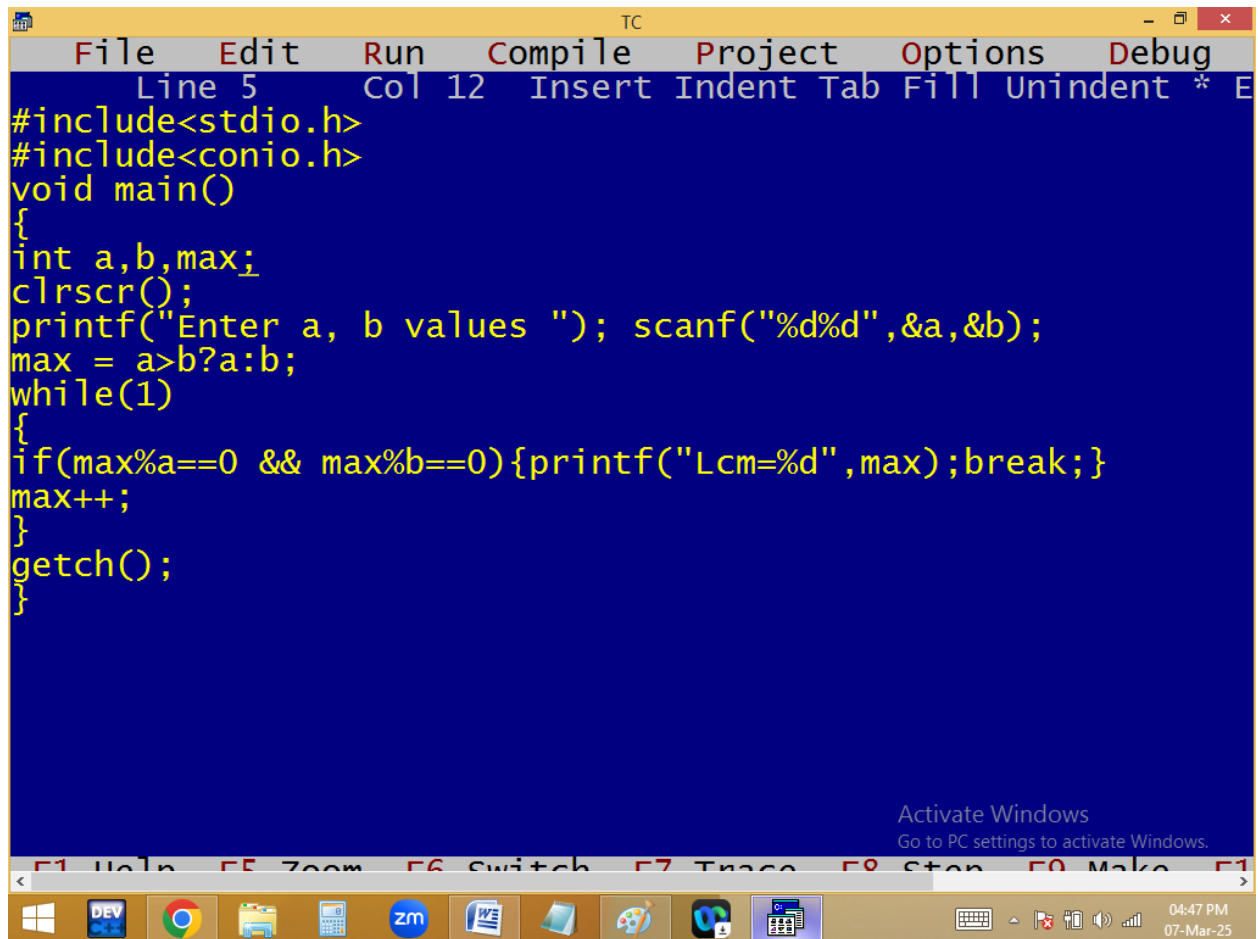
$$12 \% 4 = 0$$

max b

$$6 \quad 6$$

$$8 \% 6 = 2$$

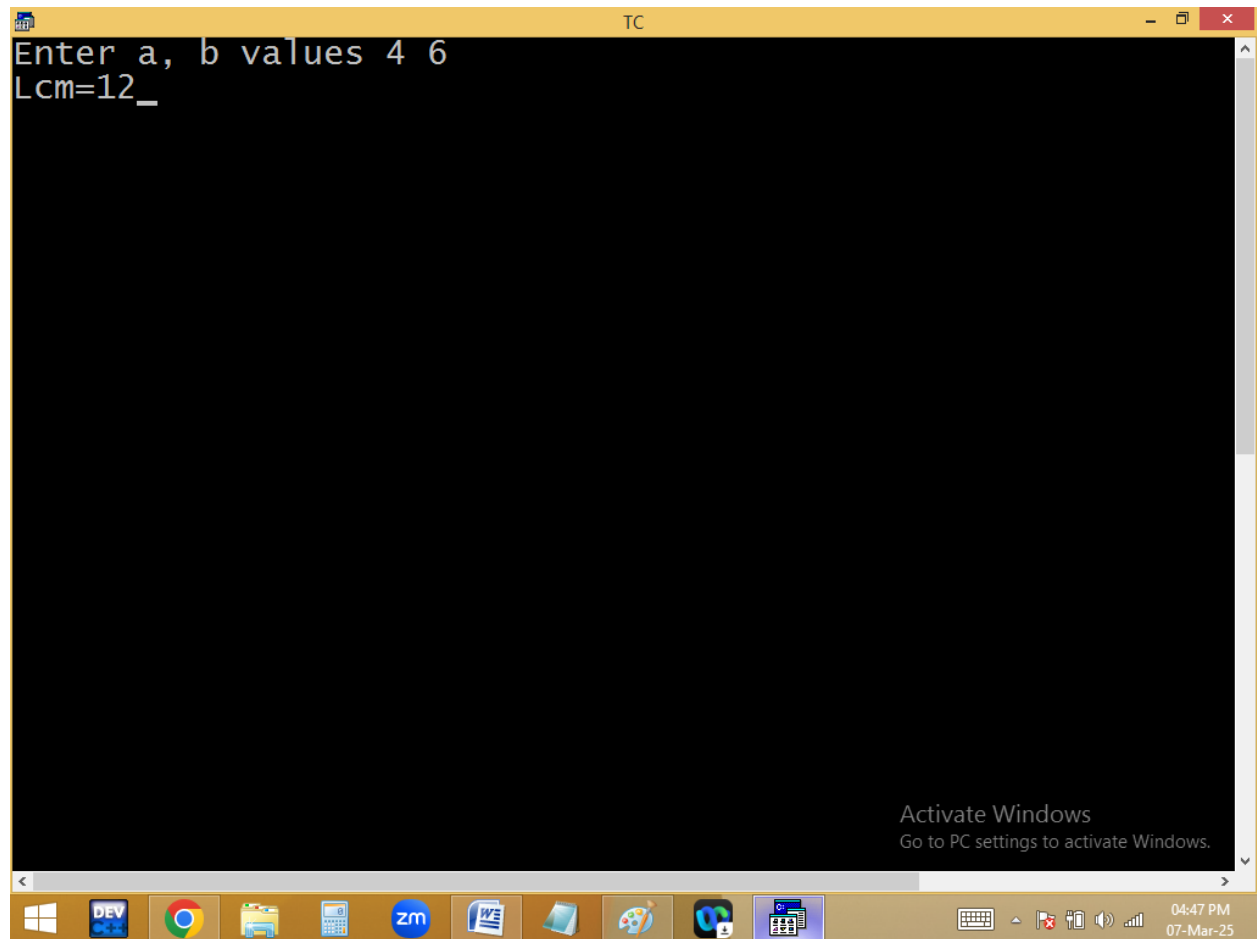
$$12 \% 6 = 0$$

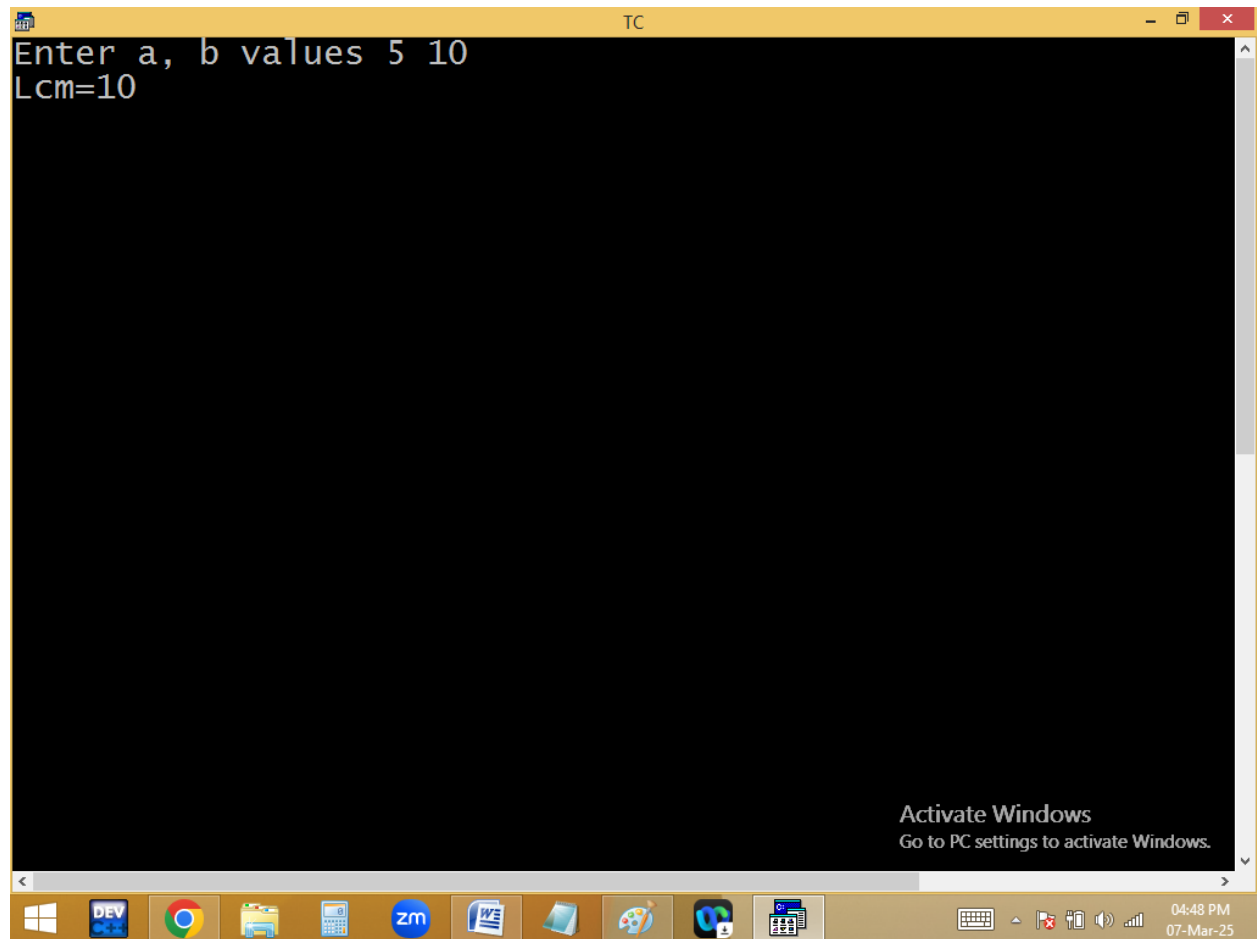


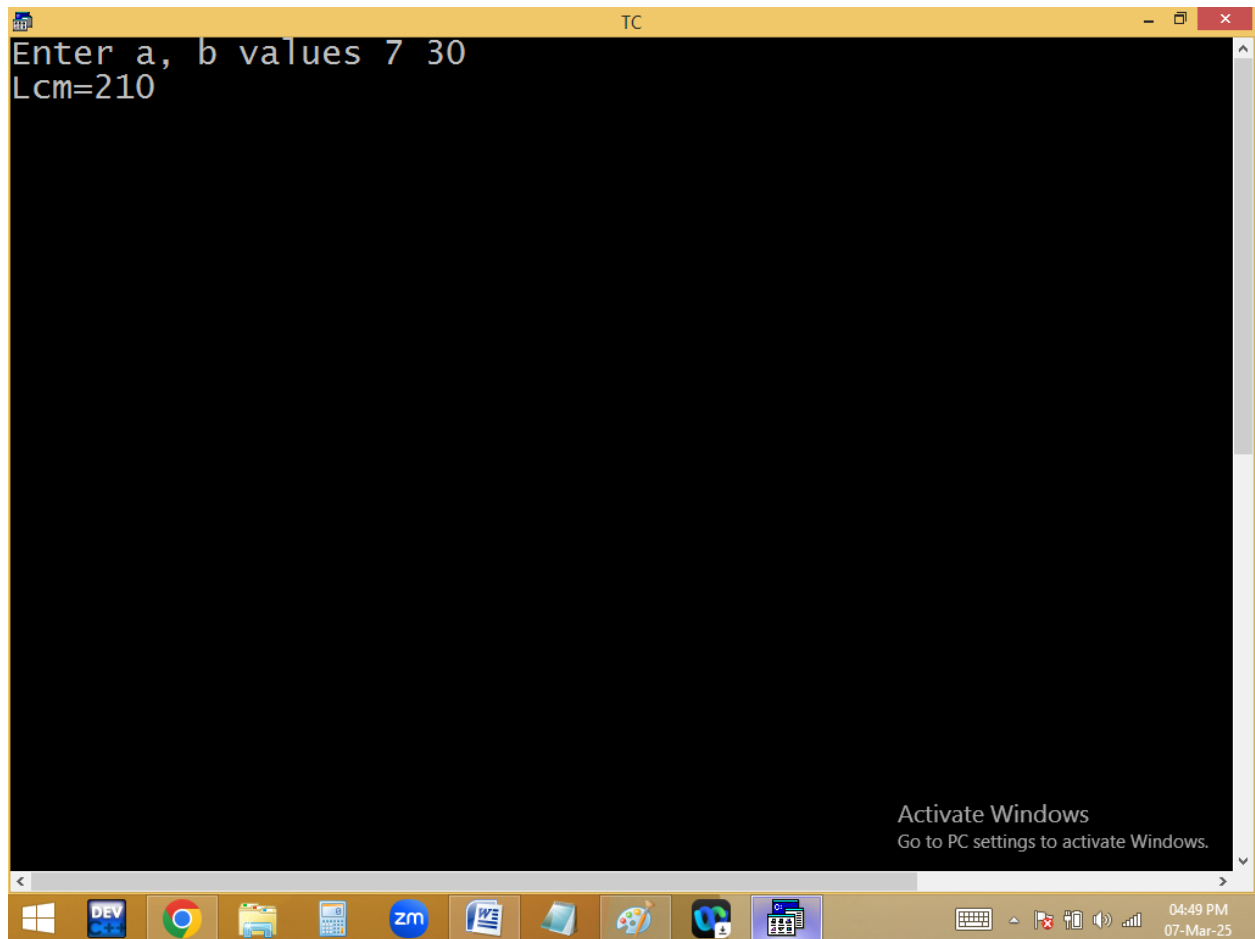
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 5" and "Col 12". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,max;
    clrscr();
    printf("Enter a, b values "); scanf("%d%d",&a,&b);
    max = a>b?a:b;
    while(1)
    {
        if(max%a==0 && max%b==0){printf("Lcm=%d",max);break;}
        max++;
    }
    getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and others. The system tray on the right shows the time as 04:47 PM and the date as 07-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







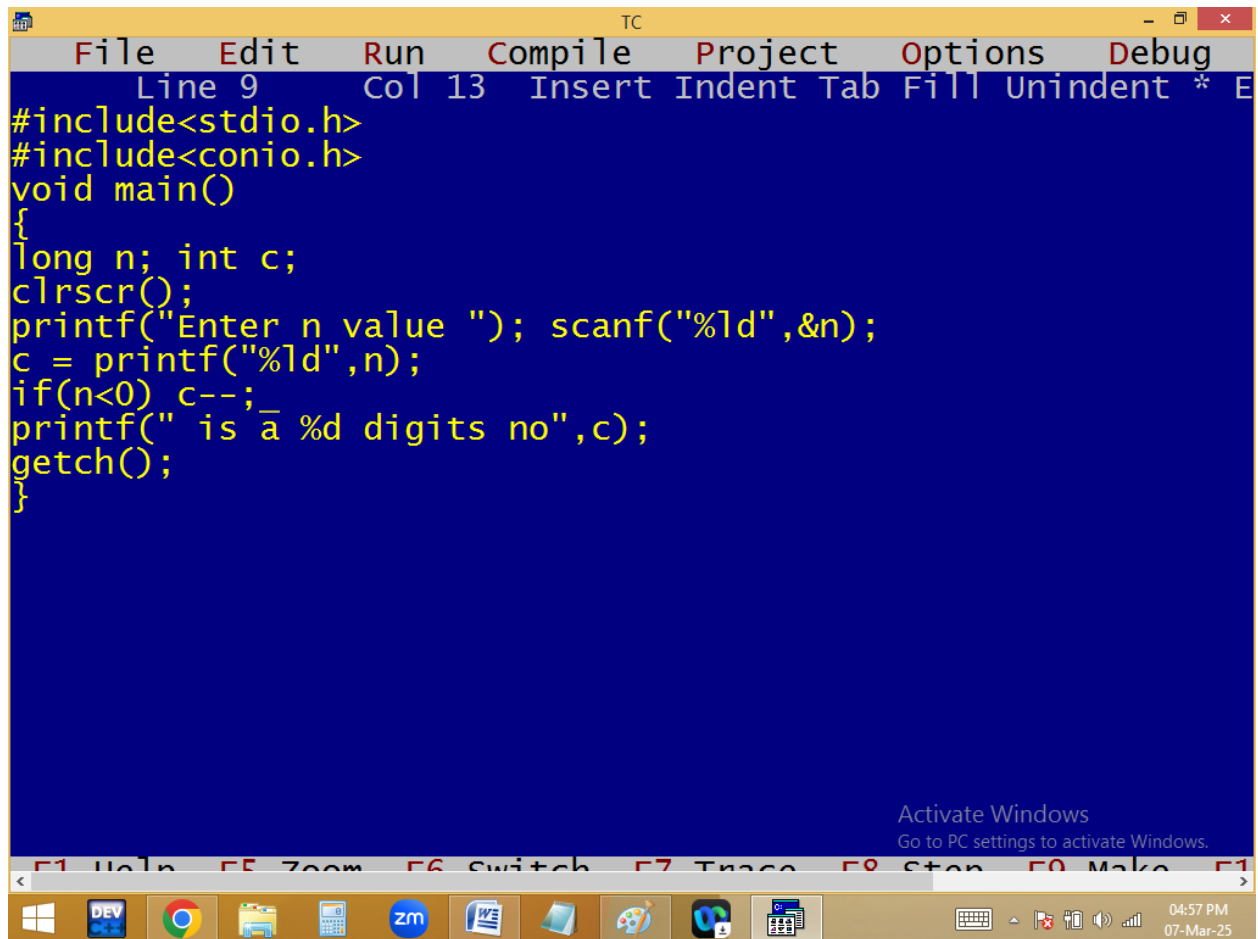
The screenshot shows a Turbo C++ (TC) window with a black background. The text "Enter a, b values 7 30" is displayed on the first line, and "Lcm=210" is on the second line. The window title bar says "TC". At the bottom right of the window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various icons including the Start button, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and others. The system clock in the bottom right corner shows "04:49 PM 07-Mar-25".

```
Enter a, b values 7 30
Lcm=210
```

Finding no of digits in given no.

Eg: 2025 = 4 digit no

Without using loop/goto label:

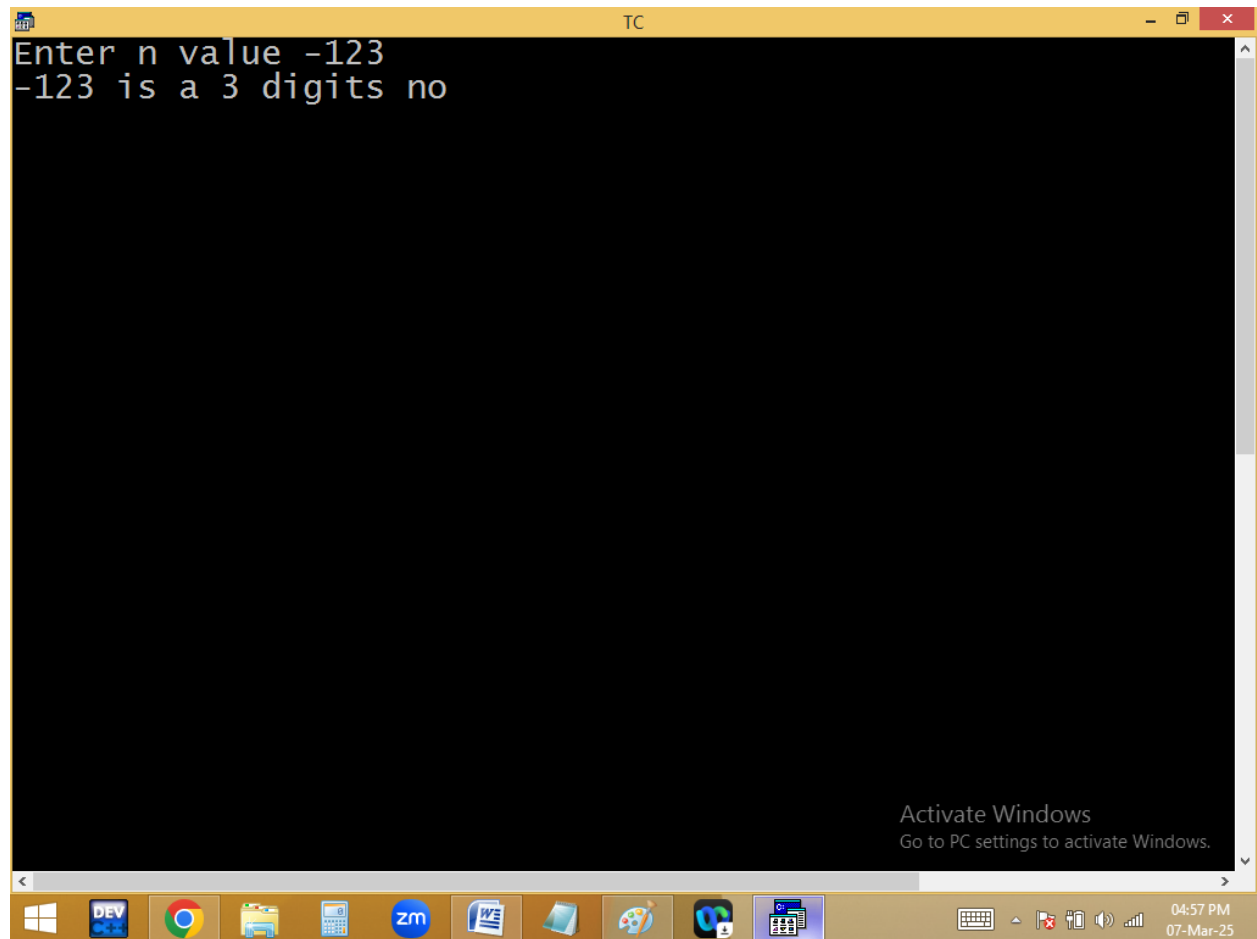


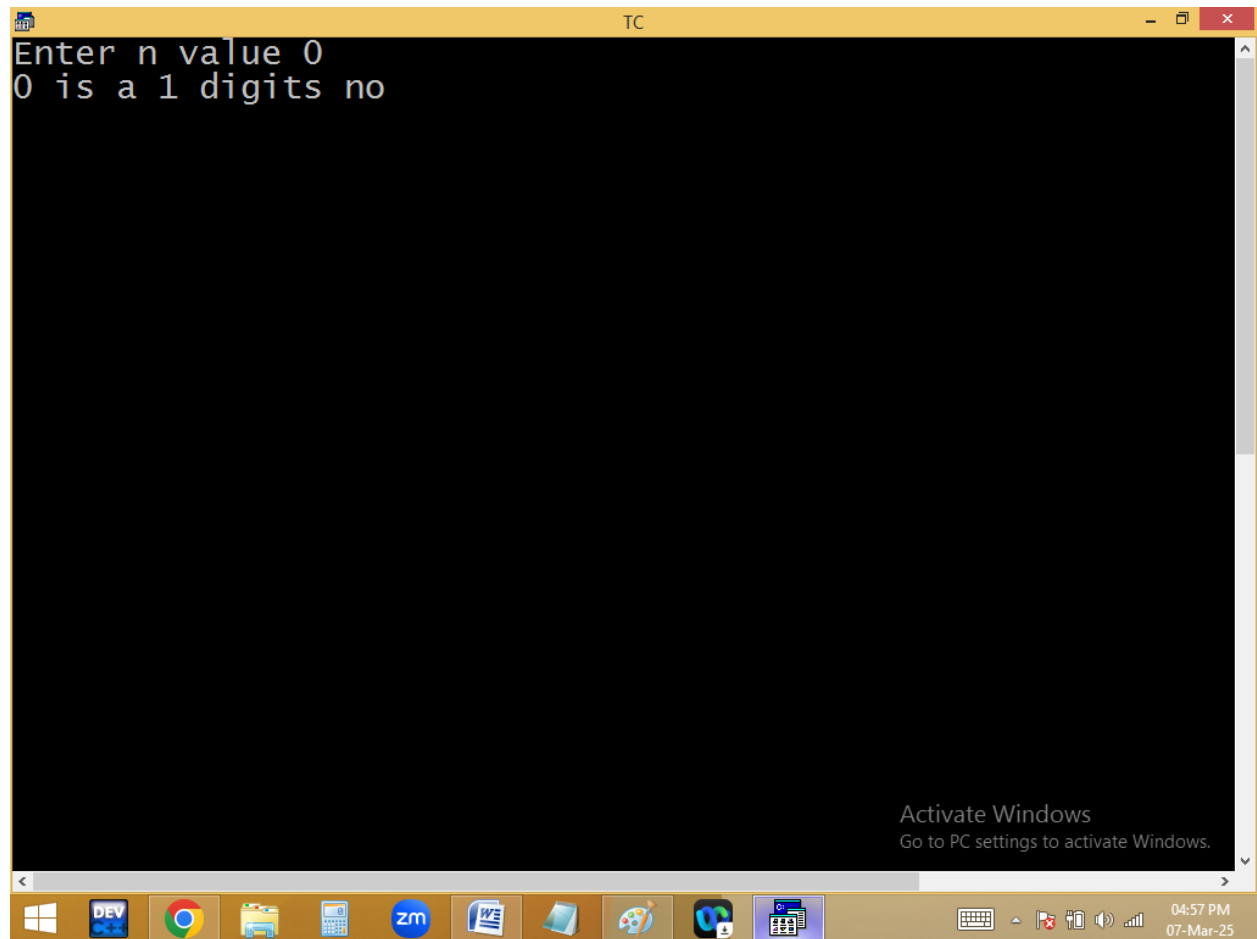
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
c = printf("%ld",n);
if(n<0) c--;
printf(" is a %d digits no",c);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows DEV Chrome File Explorer Calculator Zoom Word 04:57 PM 07-Mar-25





```
Enter n value 2025
2025 is a 4 digits no_
```

Activate Windows
Go to PC settings to activate Windows.

11
2025

4
c = printf(2025);

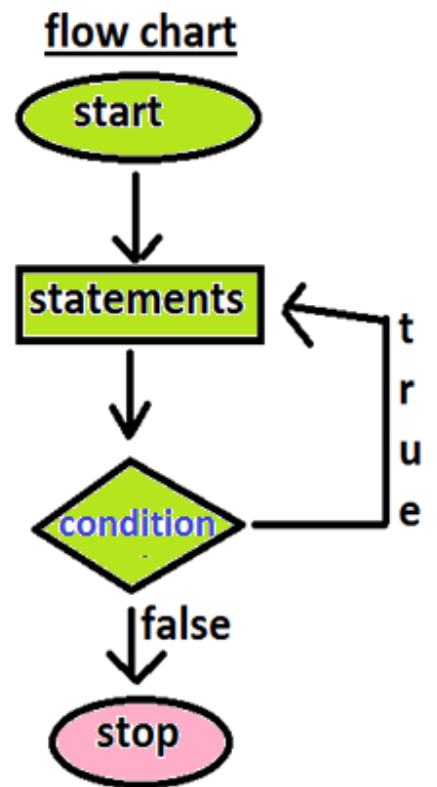
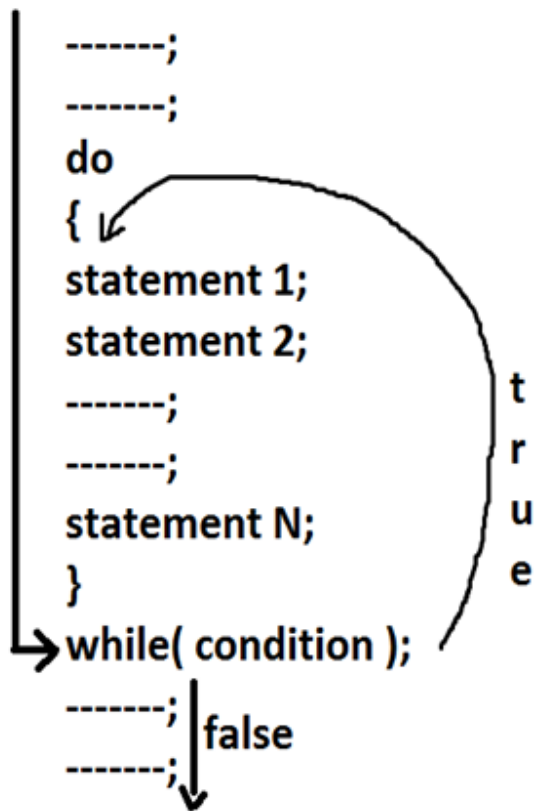
p(c);

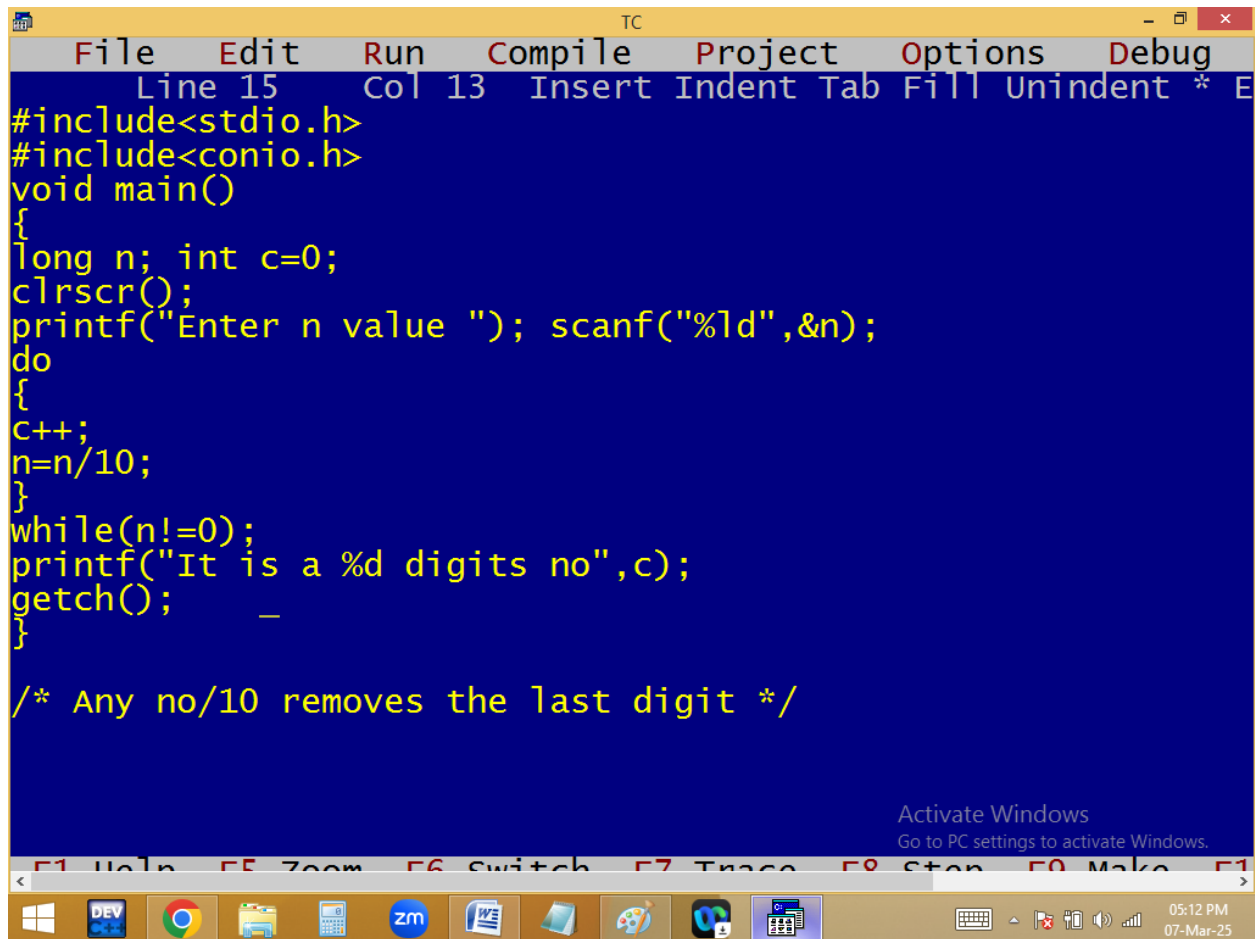
With loop:

do .. while:

- It is an exit control loop. i.e. in a do while the condition is tested at last.
- Here do , while are the keywords.
- It is also used to repeat a program several times based on a condition.
- In a do while, do block statements are executed first and later while condition is tested. If the while condition is true then once again the do block statements are repeated. Like this the process is continued until the while condition becomes false.
- In do while, the while should be end with semicolon (;) .

- **Regardless of while condition, the do statements are executed at least one time. Due to this sometimes we are getting unwanted results [garbage values].**
- **Use do while whenever it is compulsory because of in do while the program is controlled at the bottom / last.**



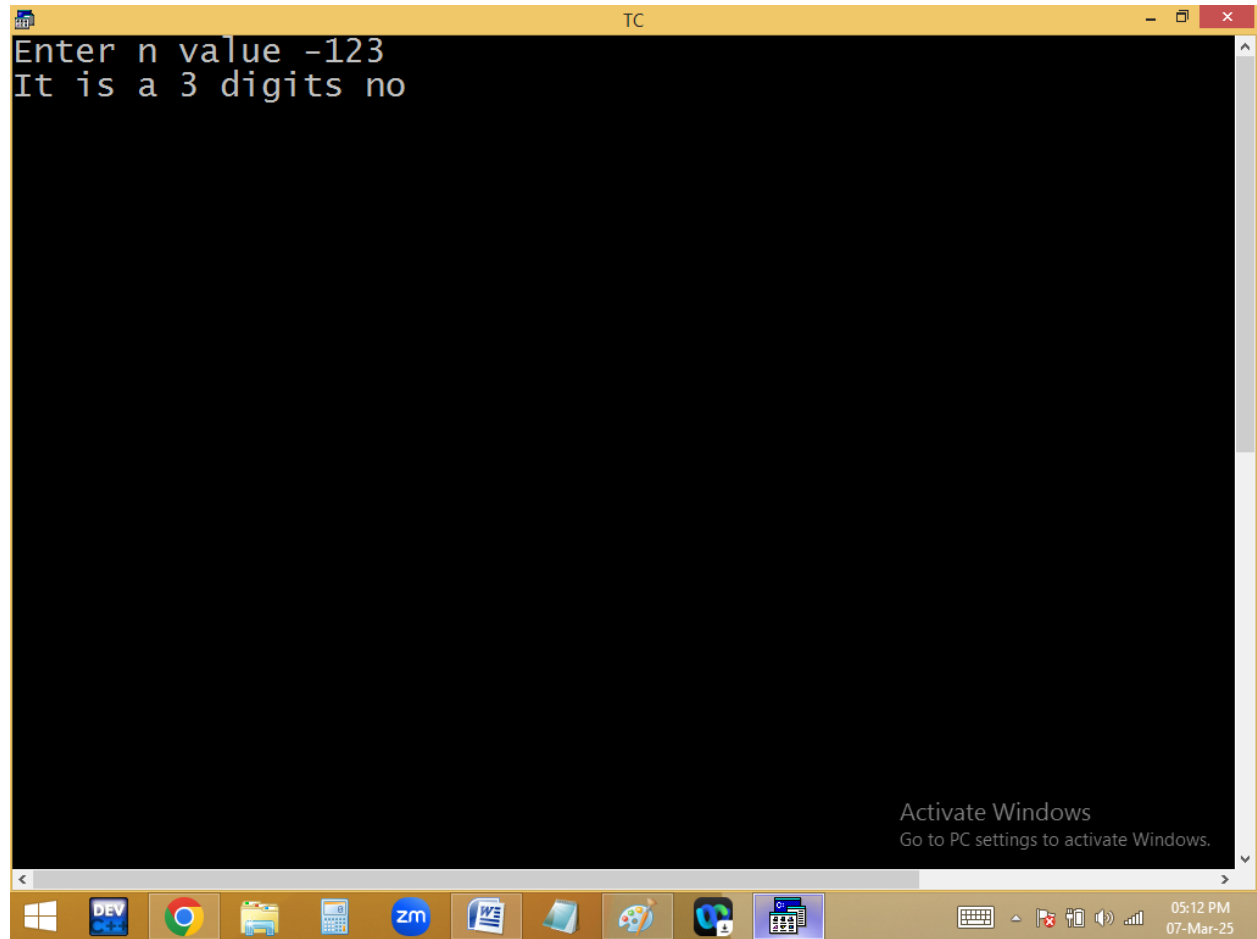


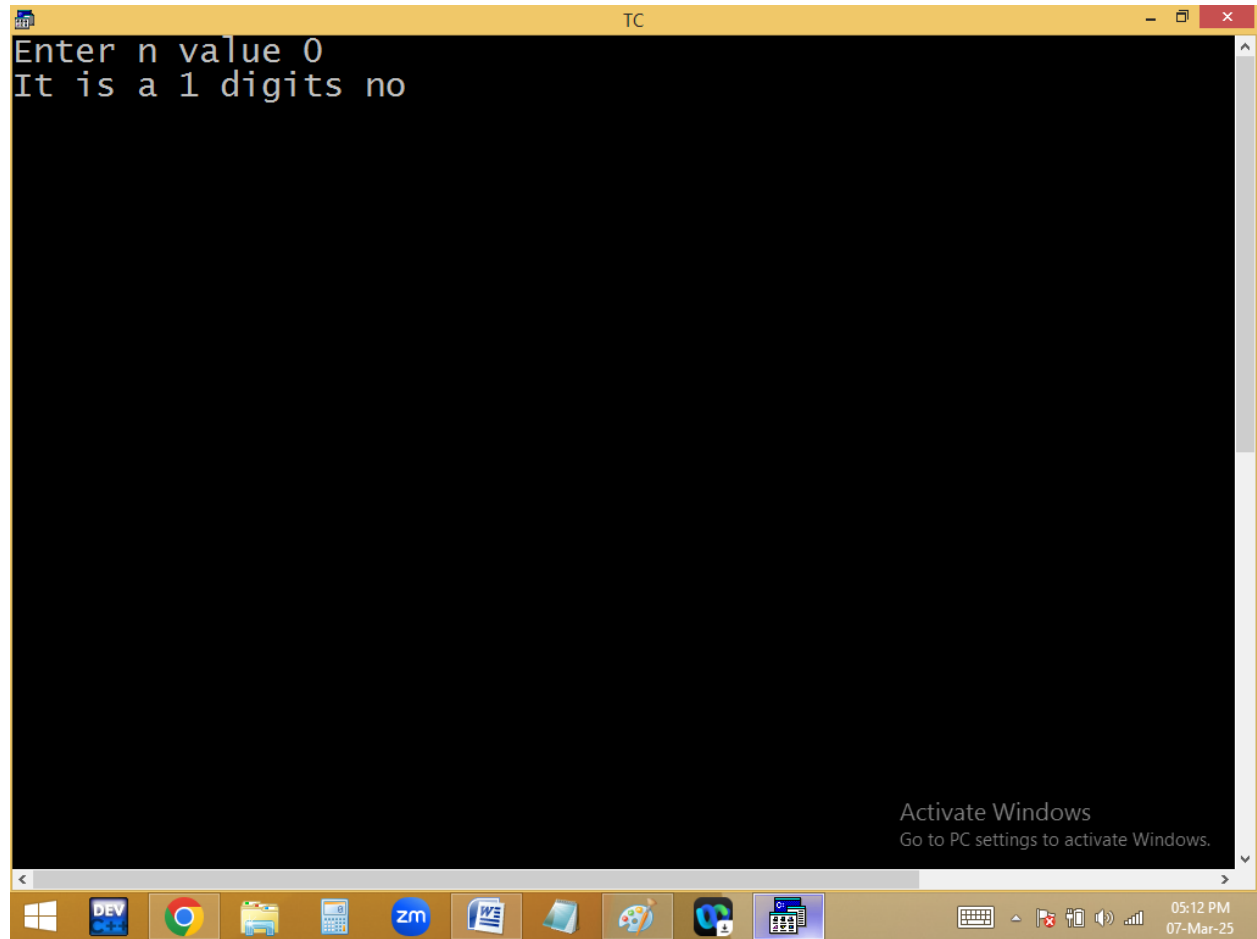
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 15" and "Col 13". The main editing area has a blue background with yellow text. The code is a C program that prompts the user to enter a number and then counts its digits by repeatedly dividing by 10. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
do
{
c++;
n=n/10;
}
while(n!=0);
printf("It is a %d digits no",c);
getch();
}

/* Any no/10 removes the last digit */
```

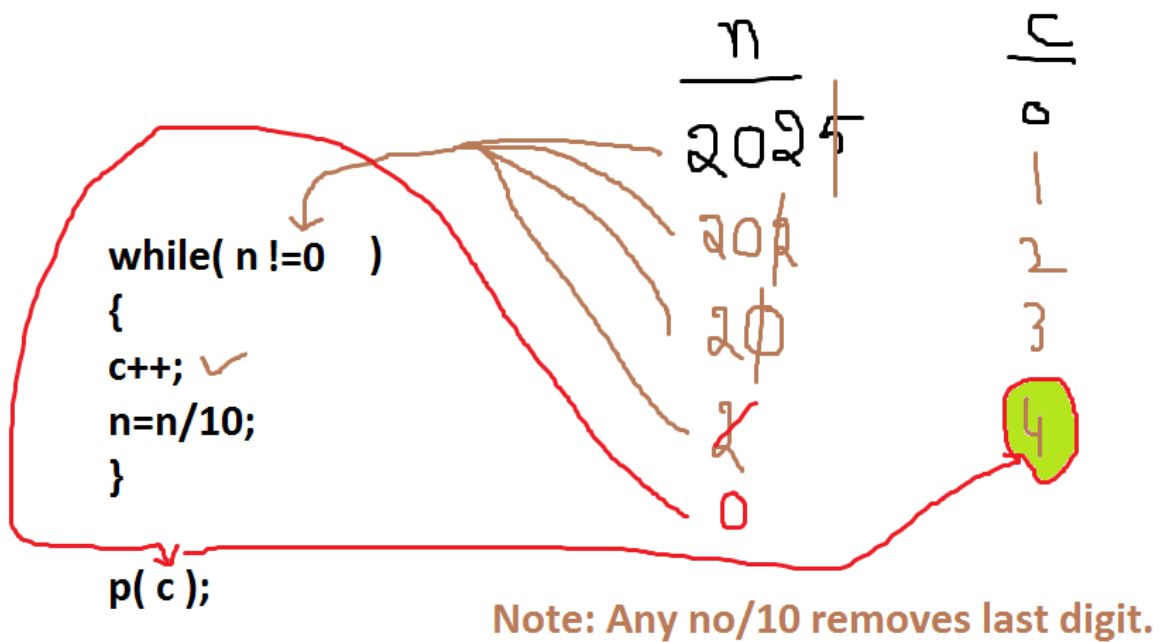
At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and others. The system tray on the right shows the time as 05:12 PM and the date as 07-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





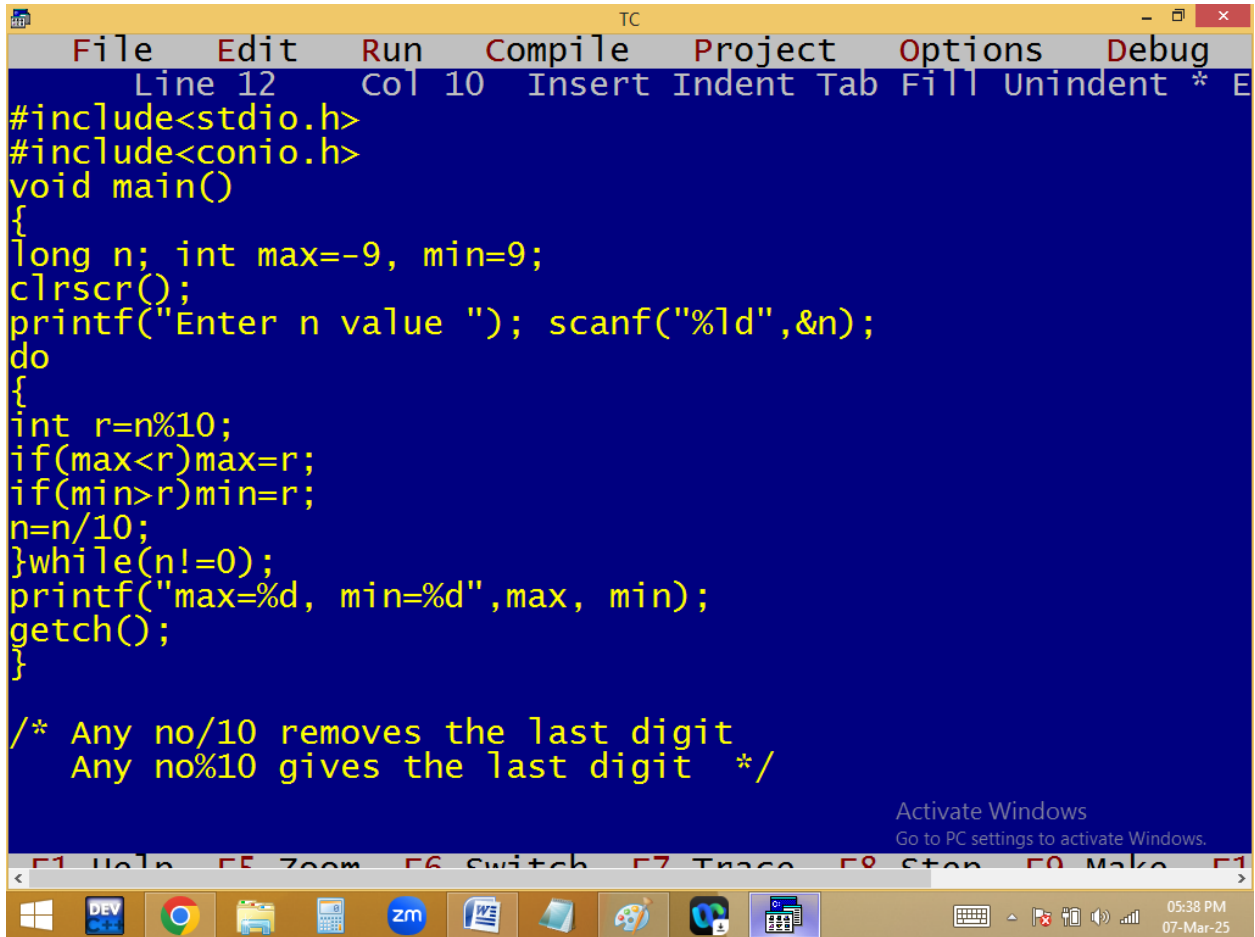
```
Enter n value -123
It is a 3 digits no
```

Activate Windows
Go to PC settings to activate Windows.



Printing the max and min digits of given no

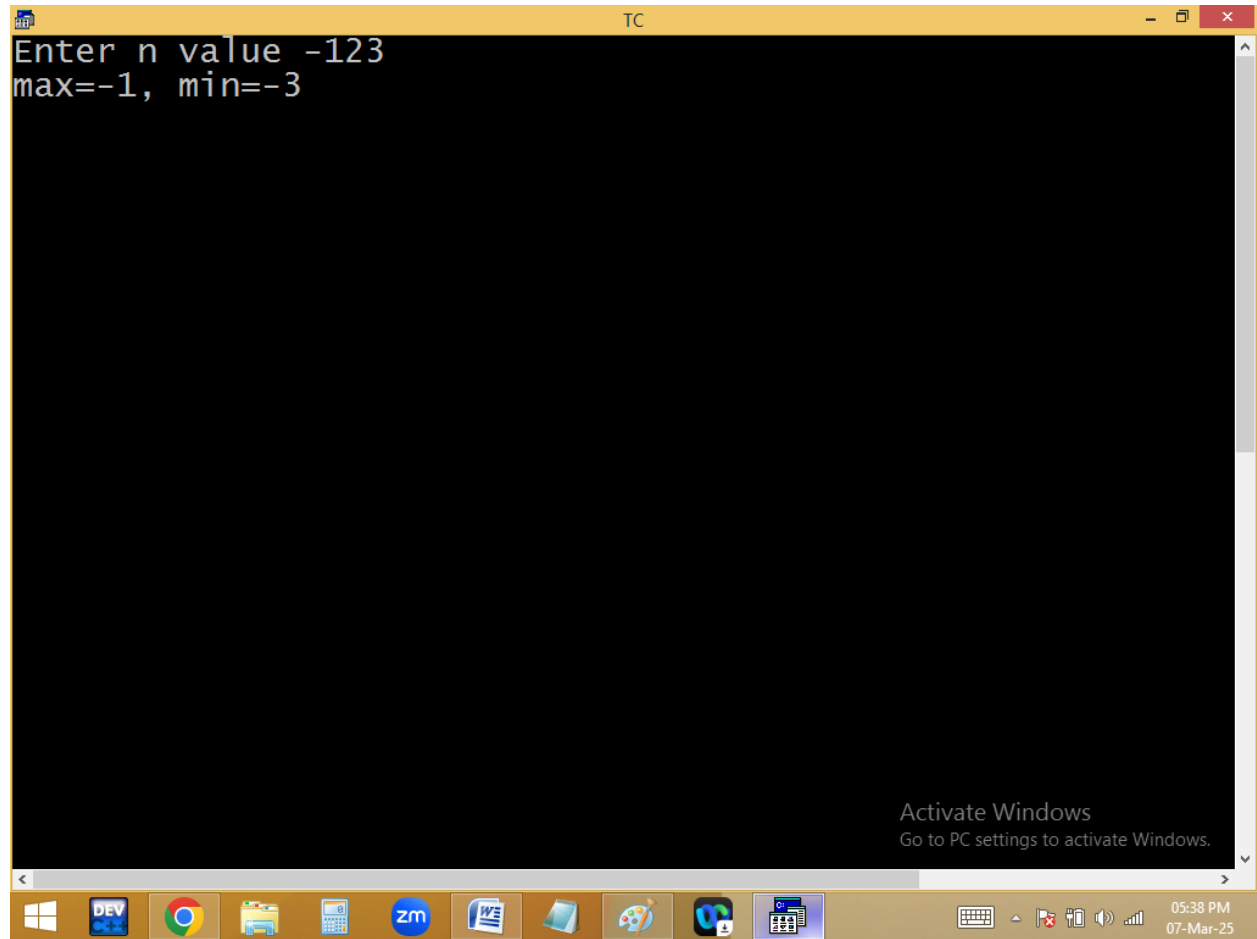
Eg: 2064 → 0-min, 6-max

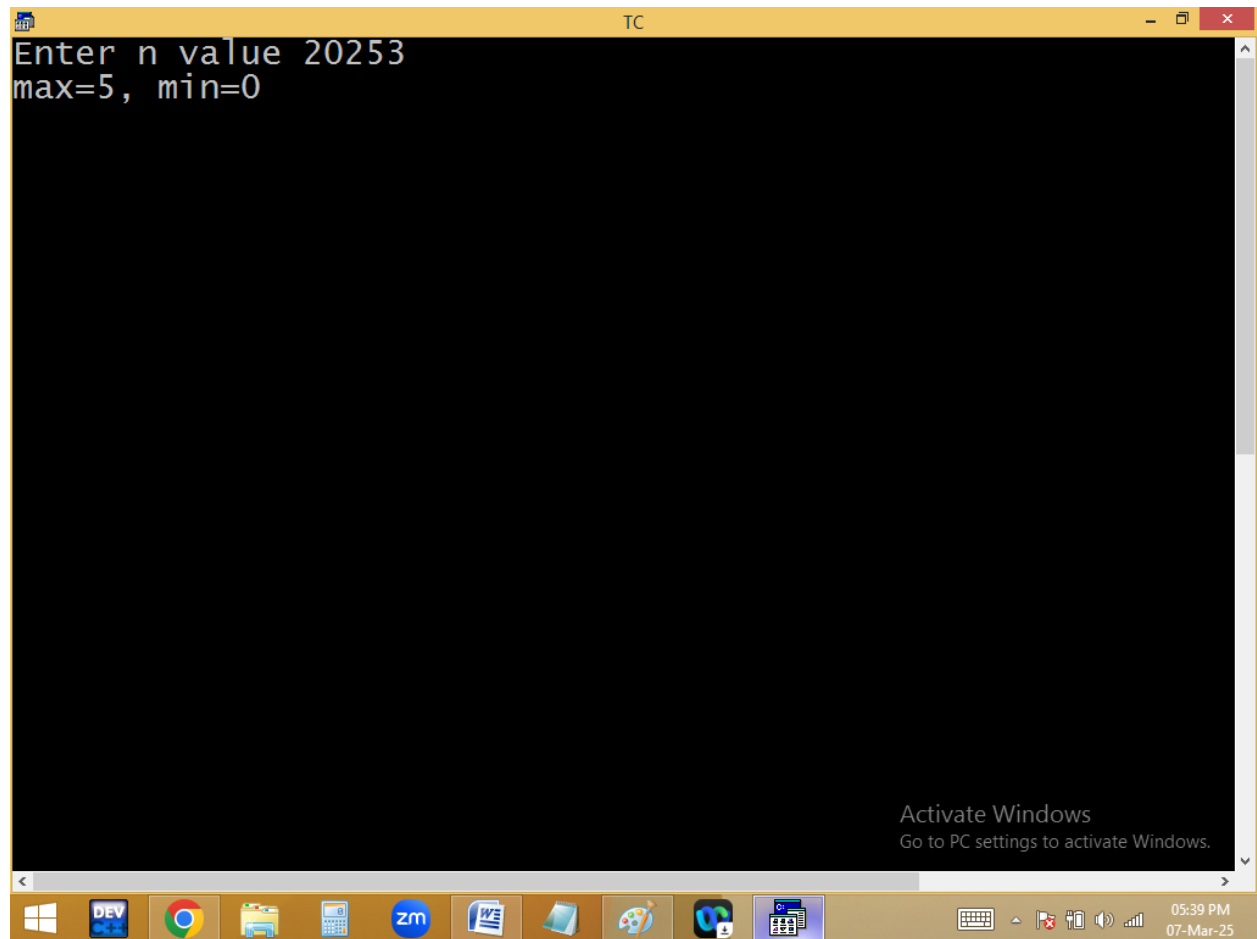


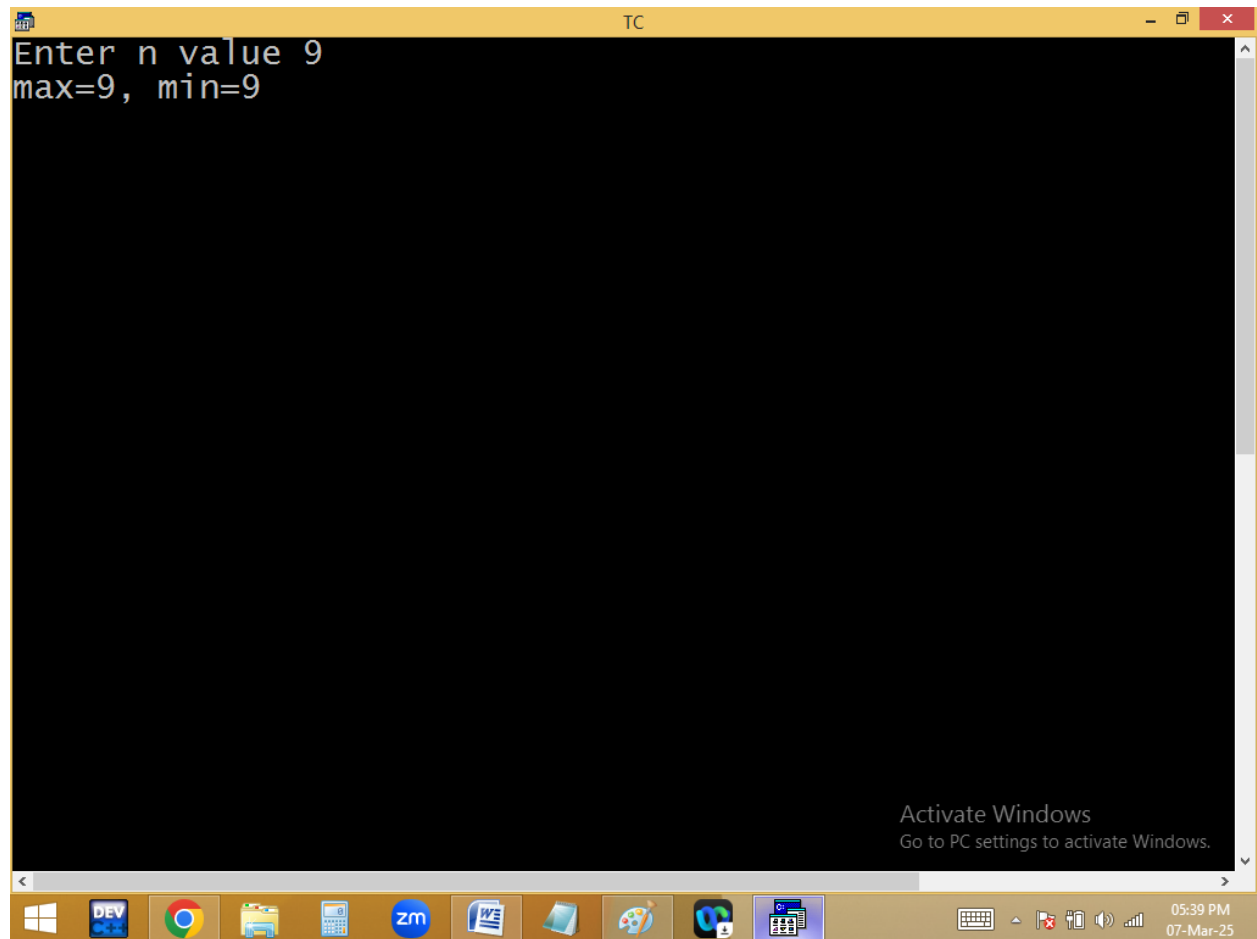
```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int max=-9, min=9;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
do
{
int r=n%10;
if(max<r)max=r;
if(min>r)min=r;
n=n/10;
}while(n!=0);
printf("max=%d, min=%d",max, min);
getch();
}

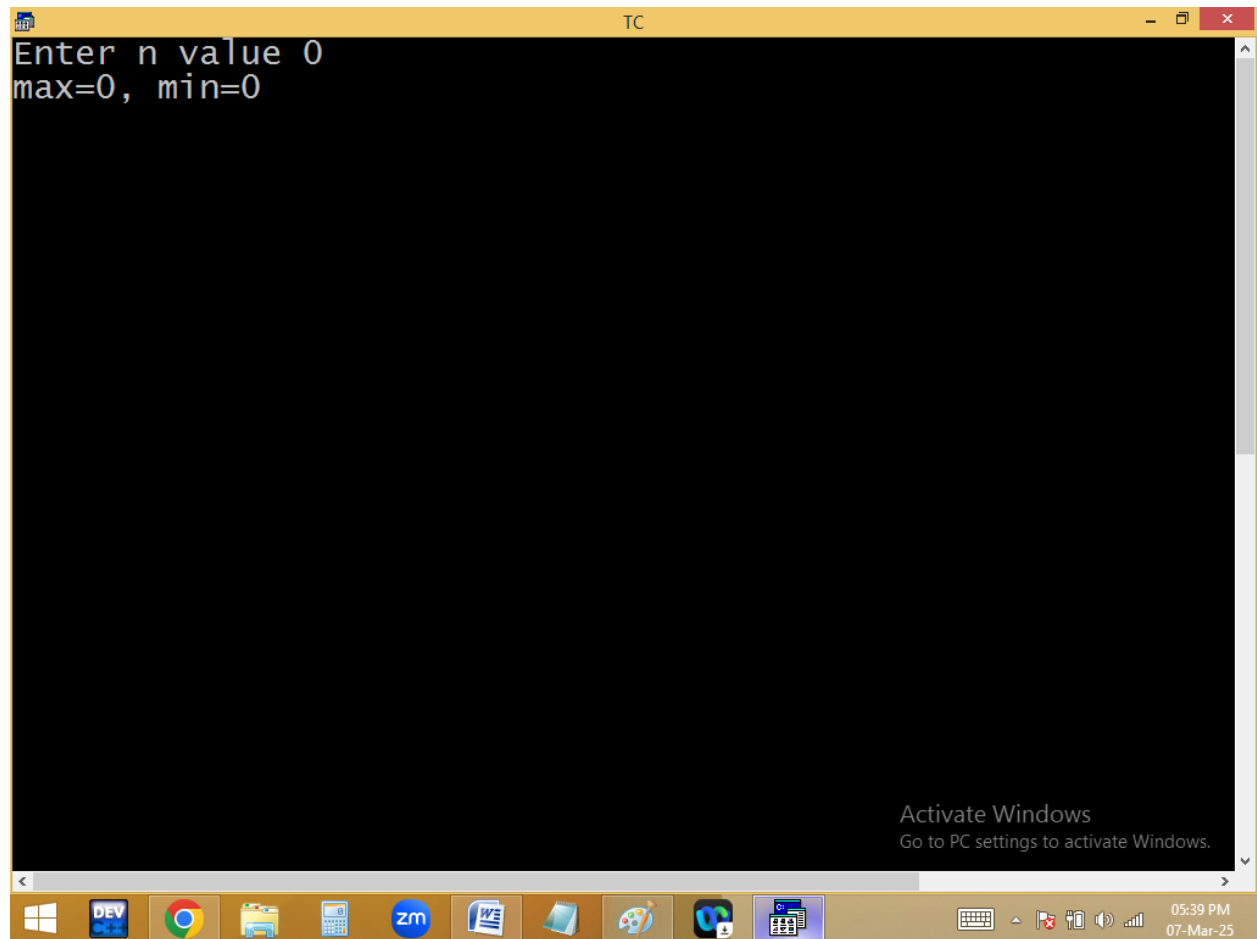
/* Any no/10 removes the last digit
Any no%10 gives the last digit */

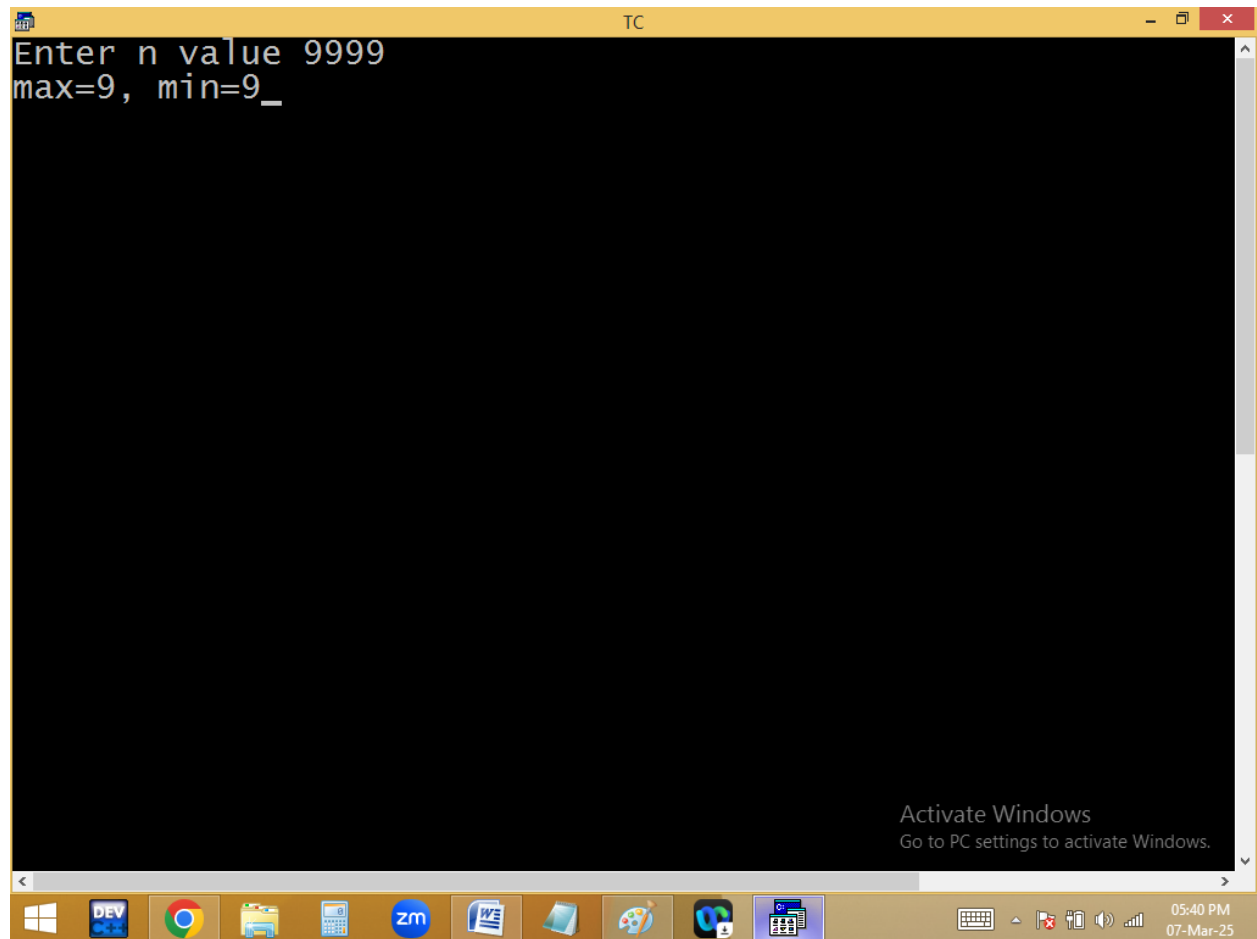
Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

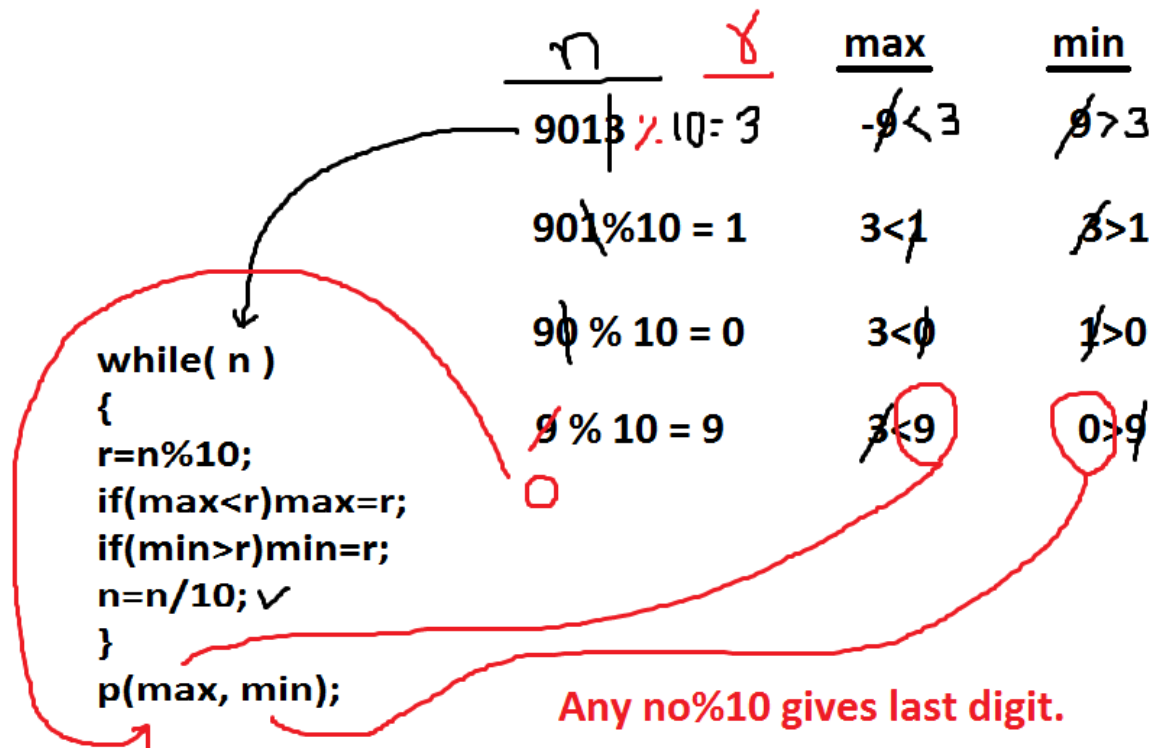






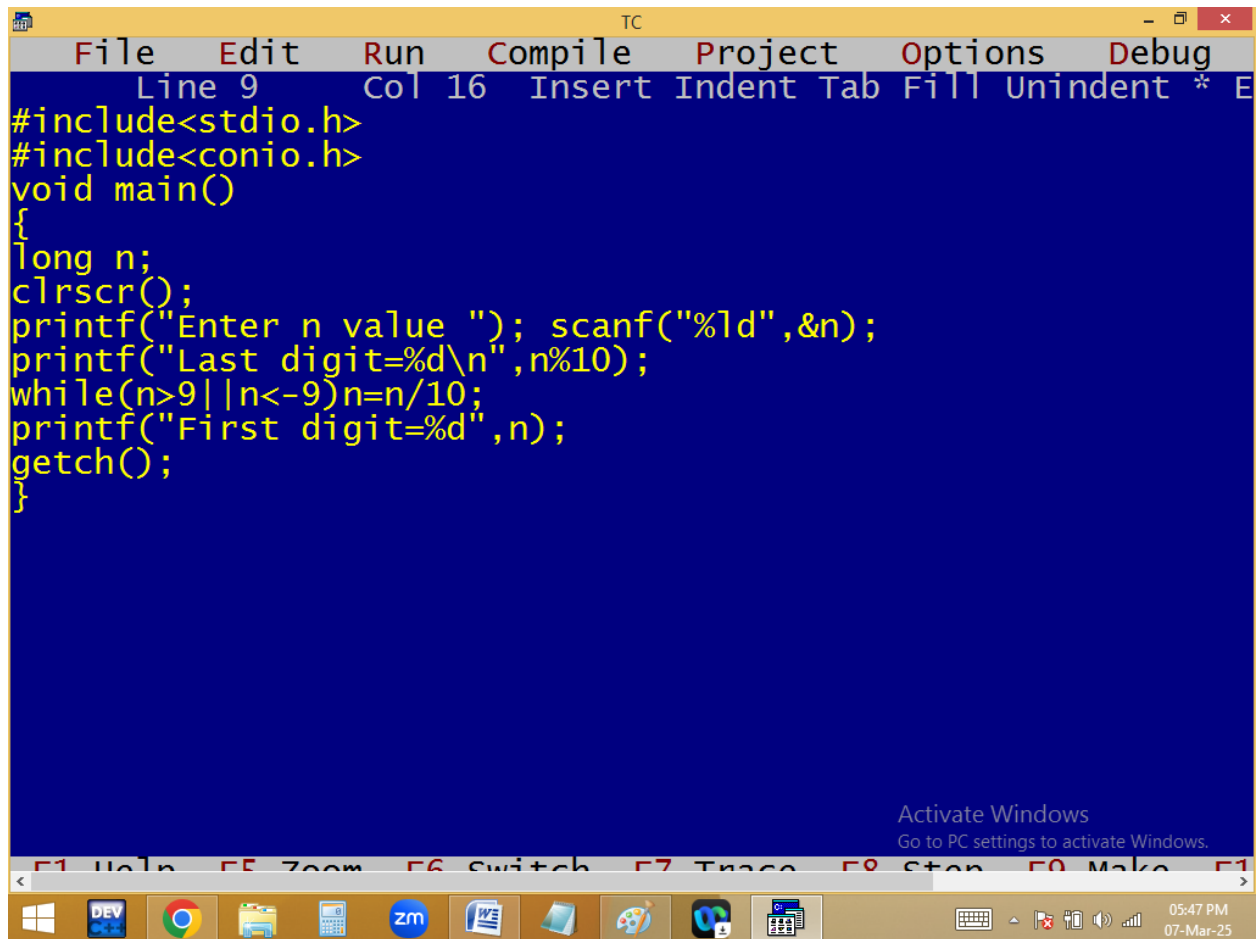






Finding first and last digits of given no?

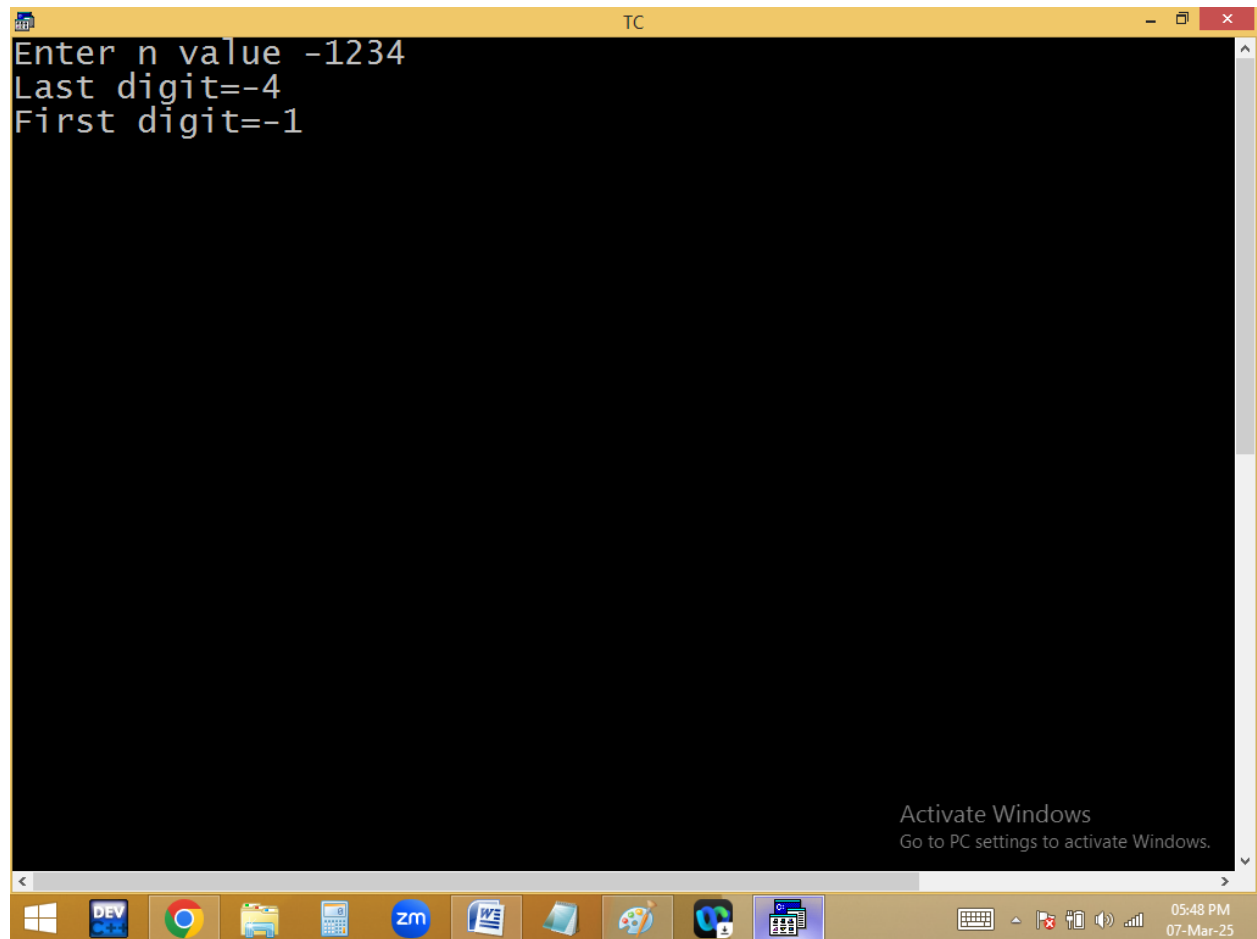
Eg: 2045 → 5 last digit, 2 first digit.

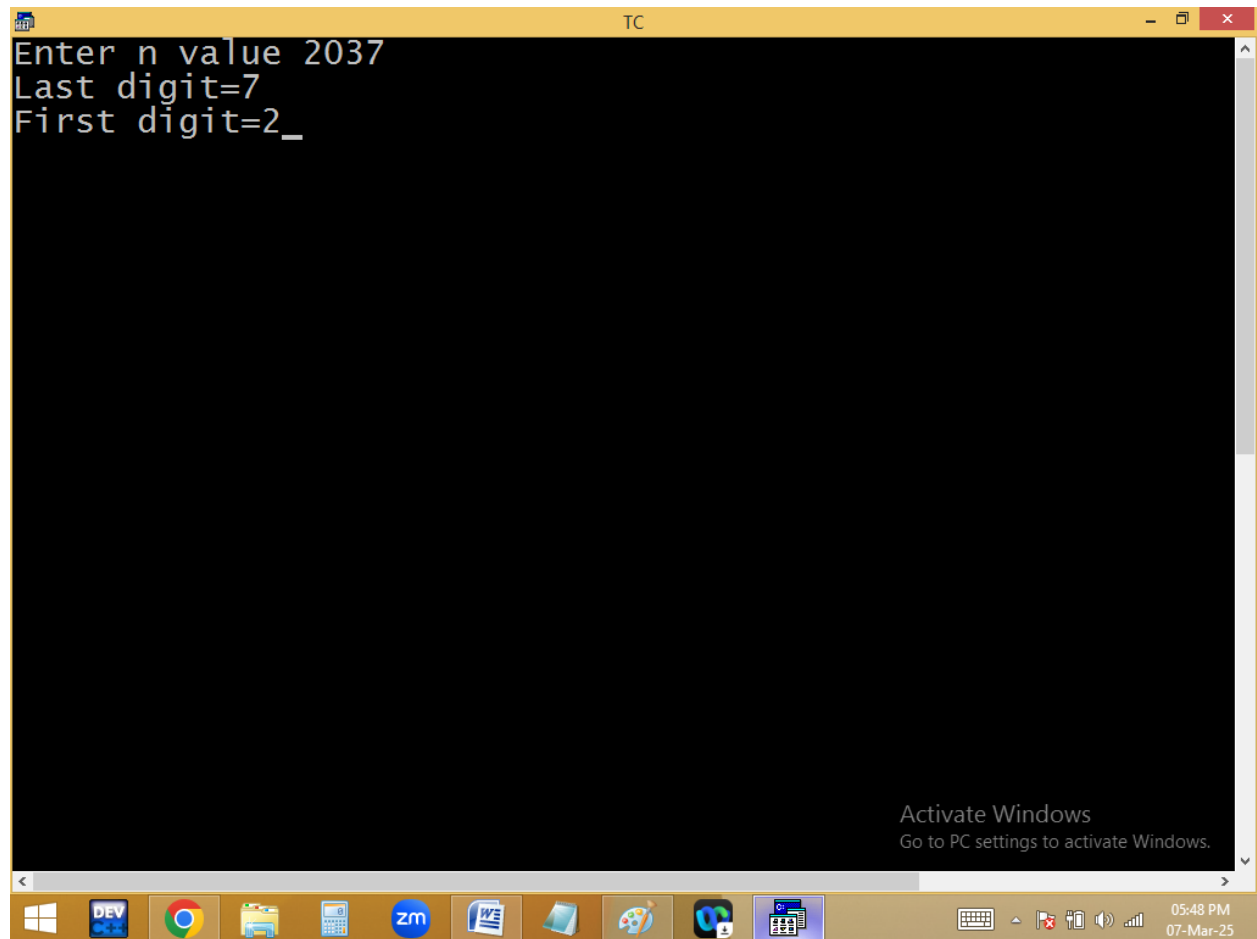


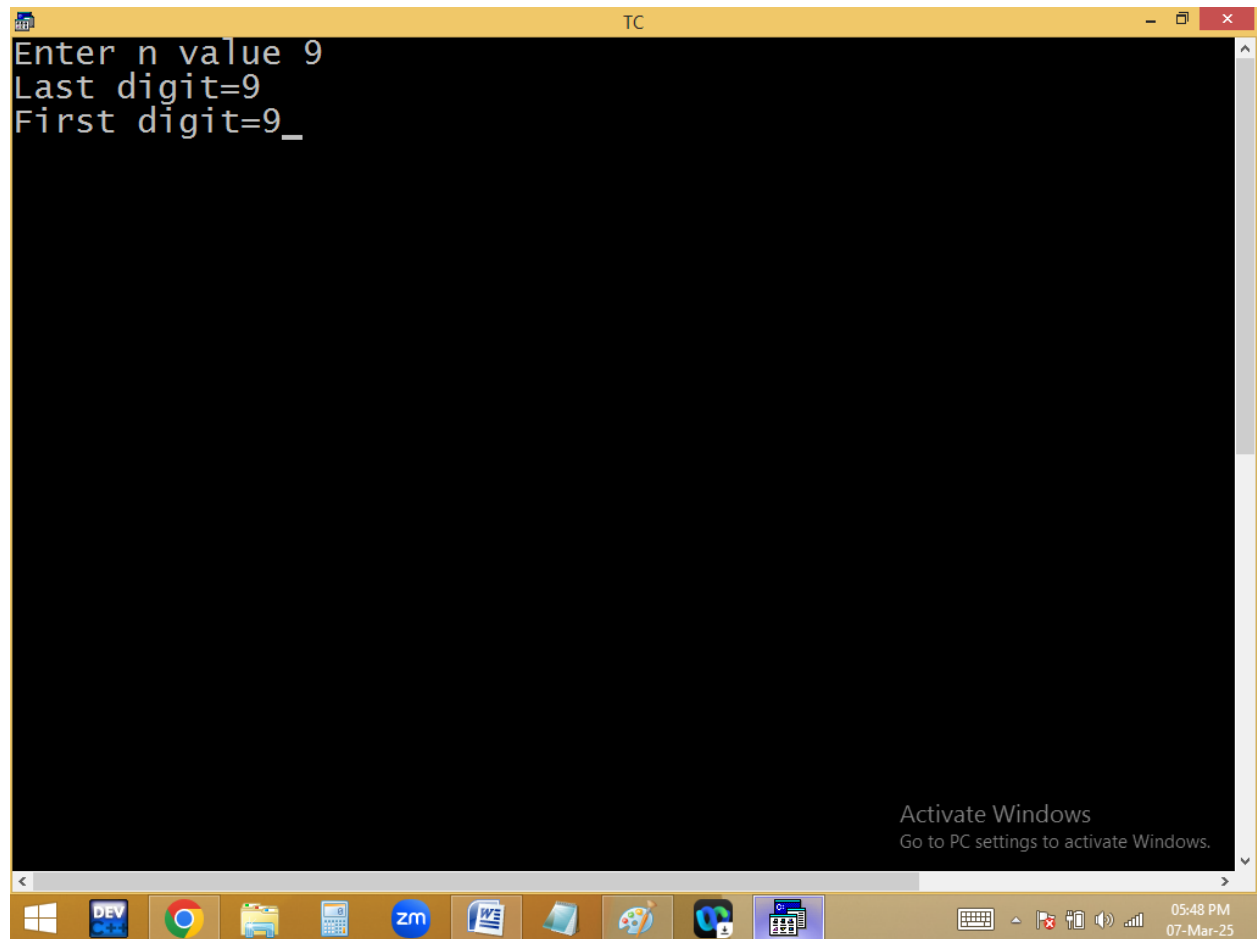
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 9", "Col 16", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

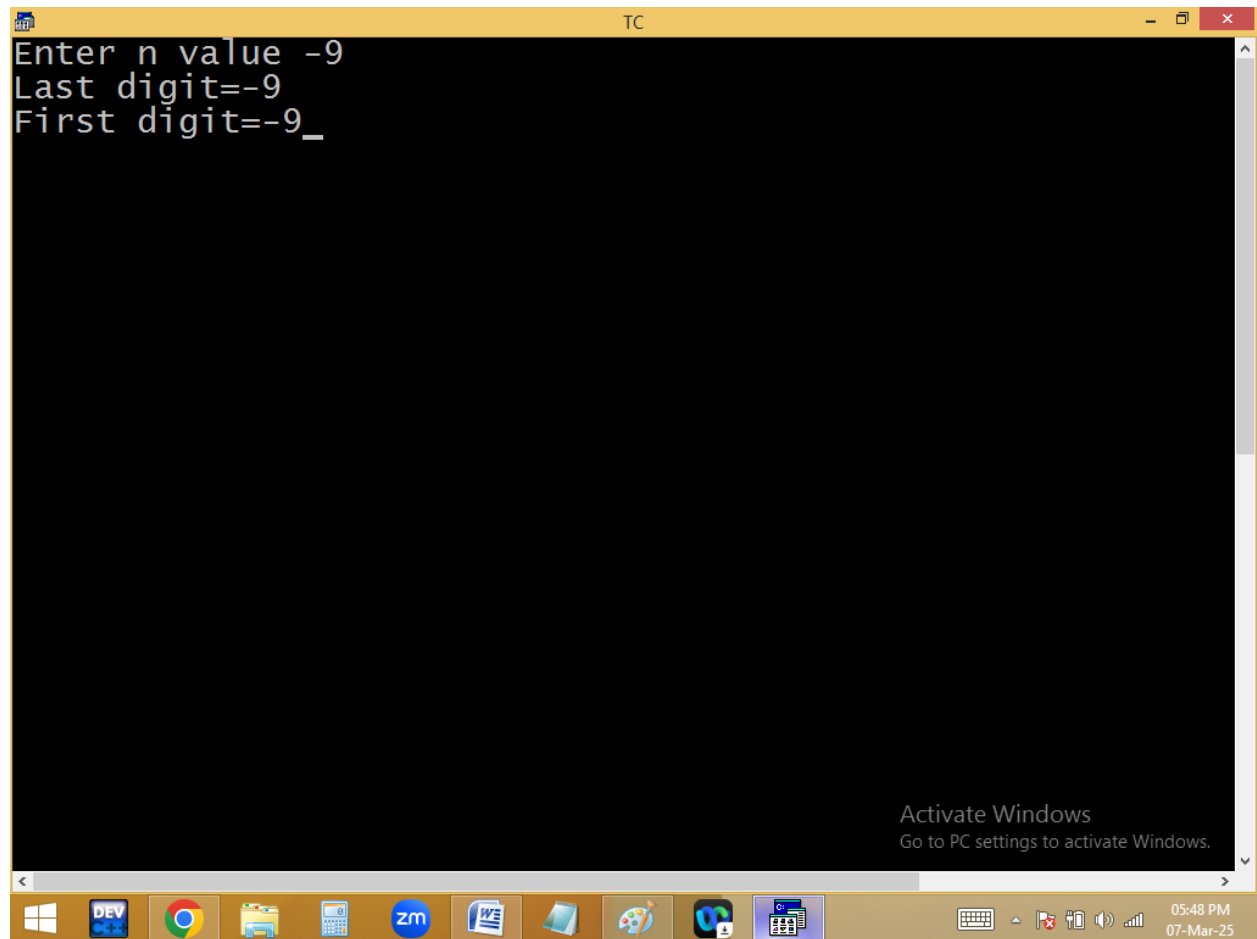
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    printf("Last digit=%d\n",n%10);
    while(n>9||n<-9)n=n/10;
    printf("First digit=%d",n);
    getch();
}
```

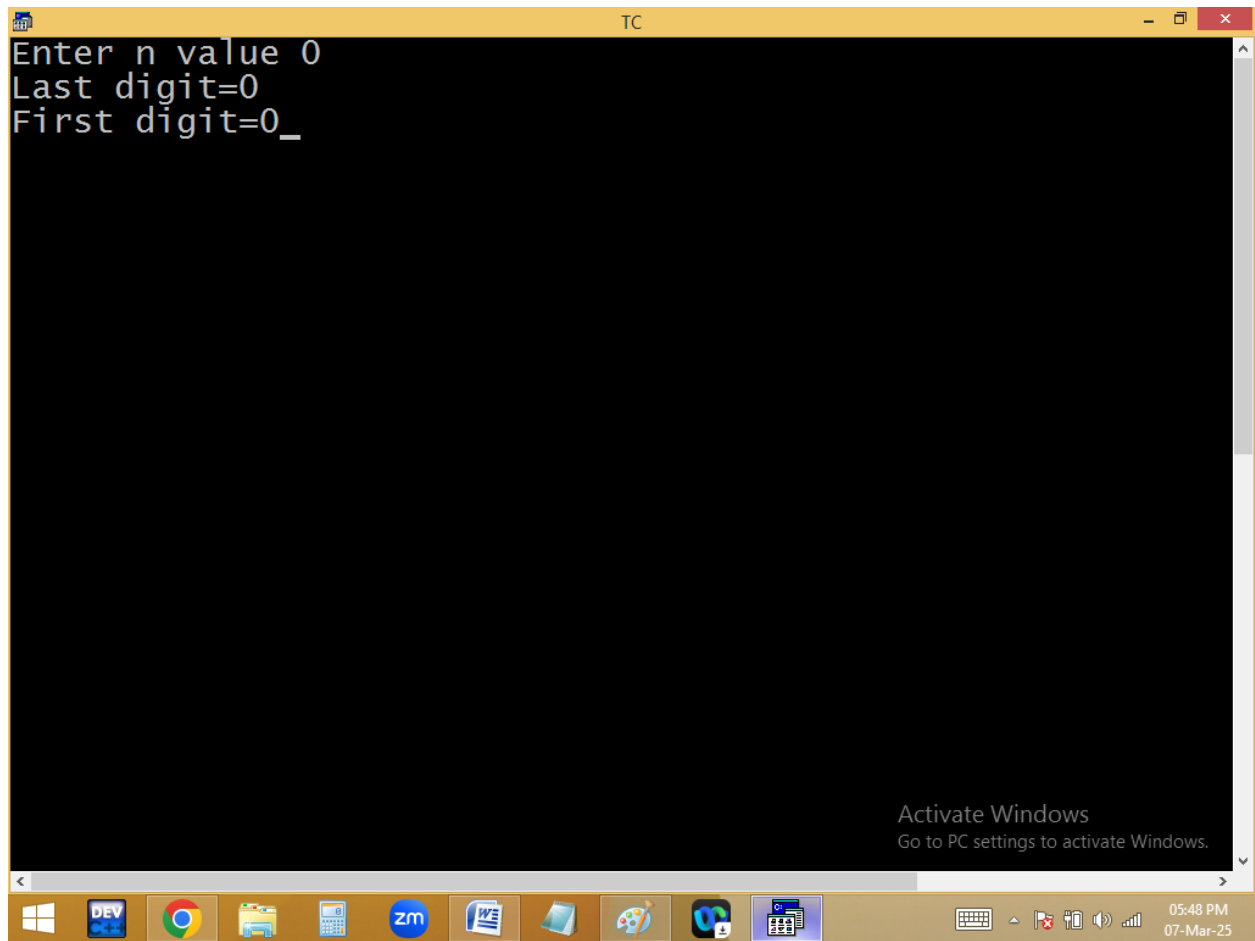
At the bottom of the window, there is a taskbar with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. A system tray on the right shows the time as 05:47 PM and the date as 07-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







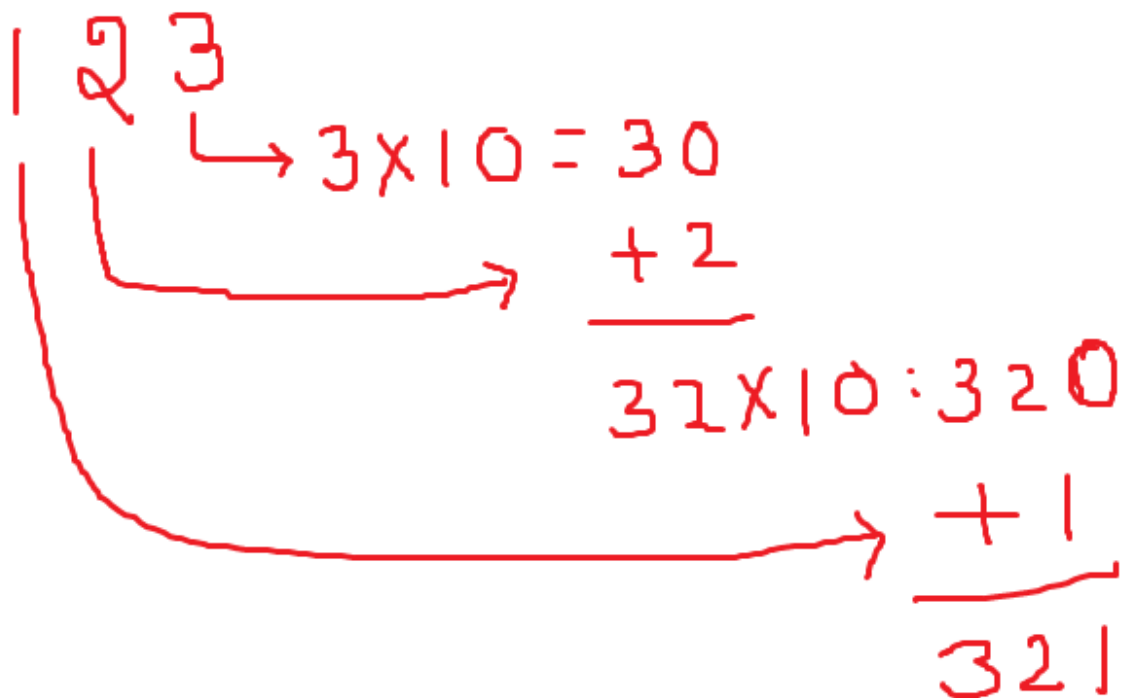




```
Enter n value 0
Last digit=0
First digit=0_
```

Activate Windows
Go to PC settings to activate Windows.

Reverse no:



100 reverse 001

12000 reverse 00021

-100 reverse is -001

2. palindrome → 101 reverse 101

3. 102 → One Zero Two

